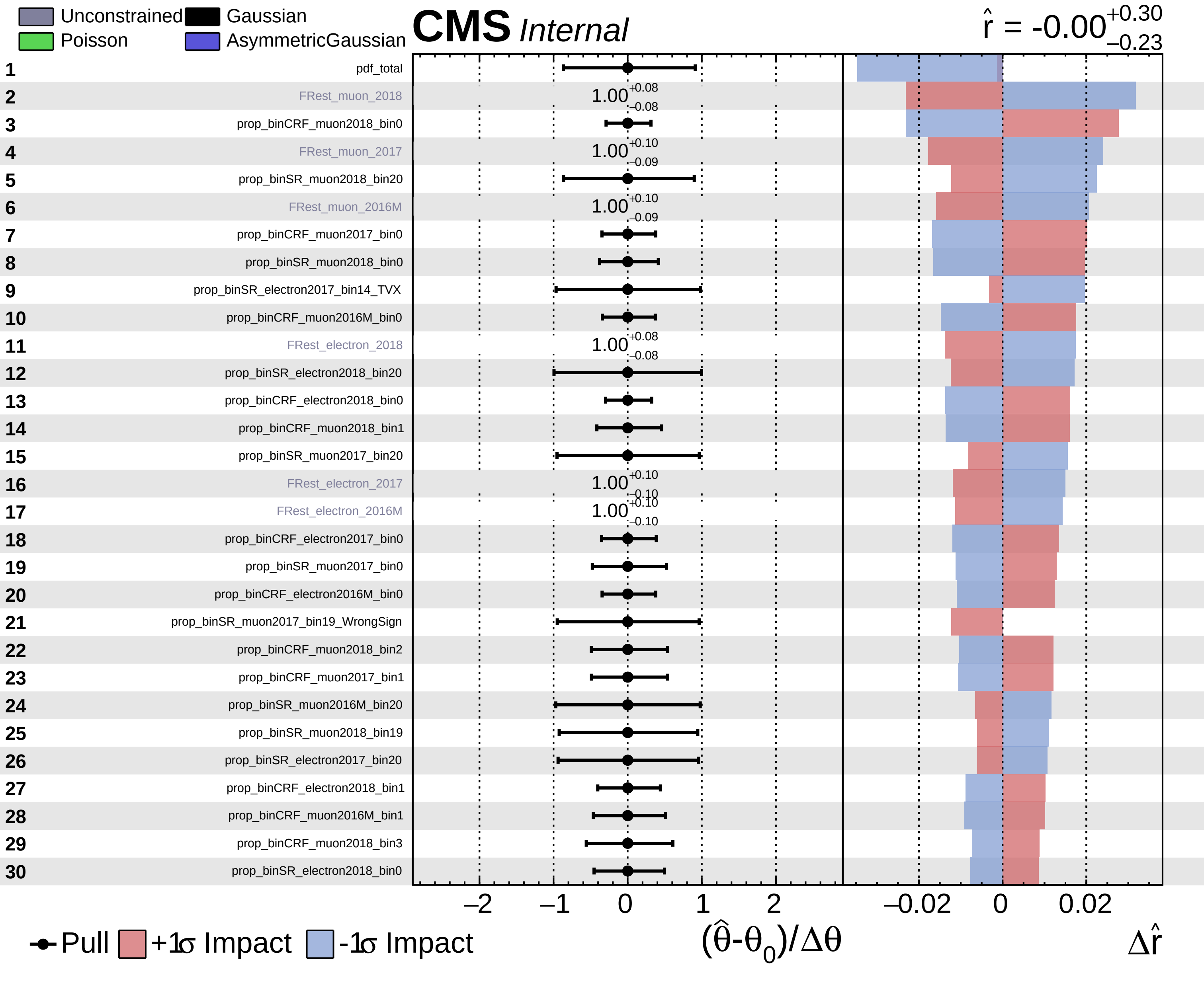
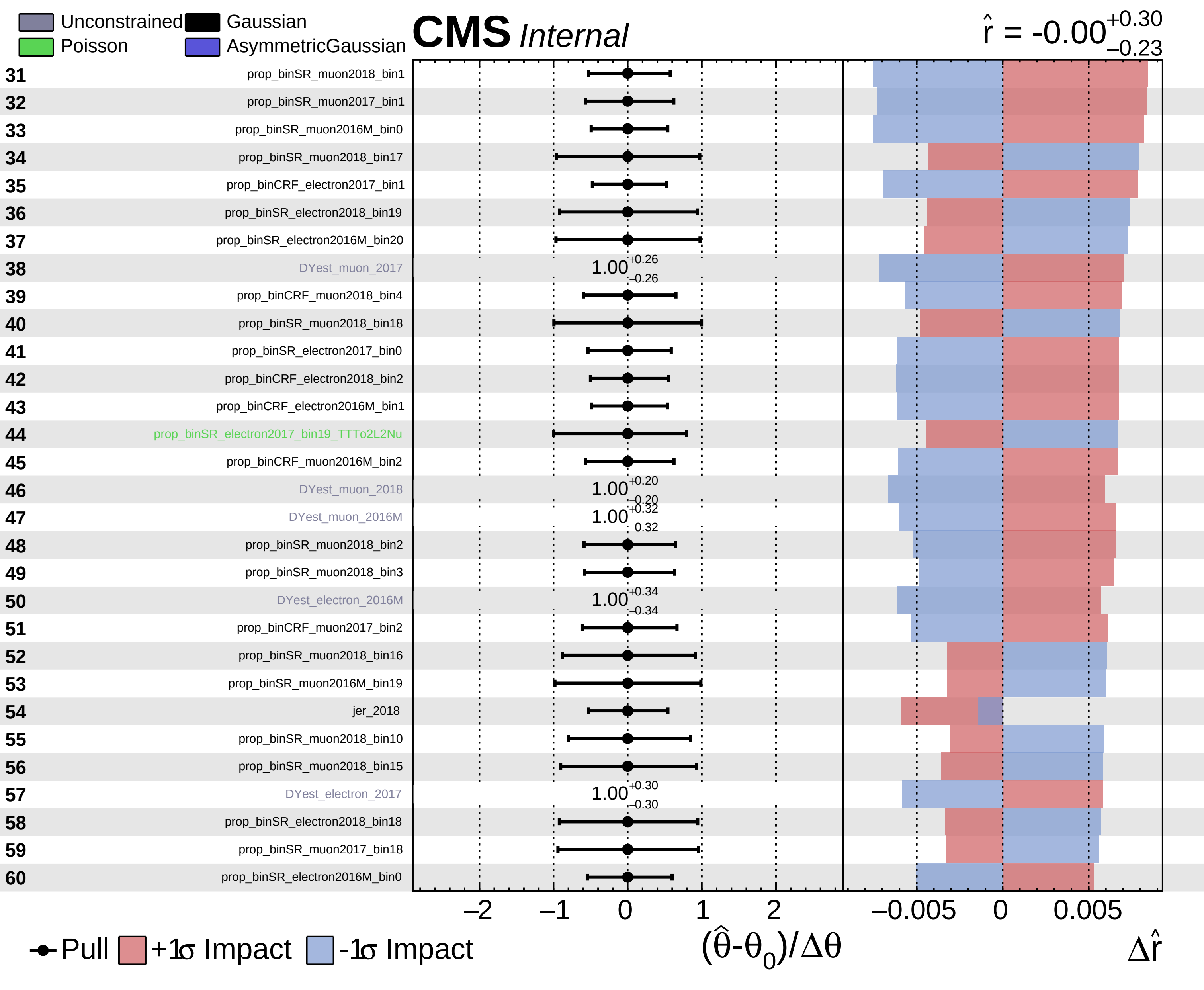
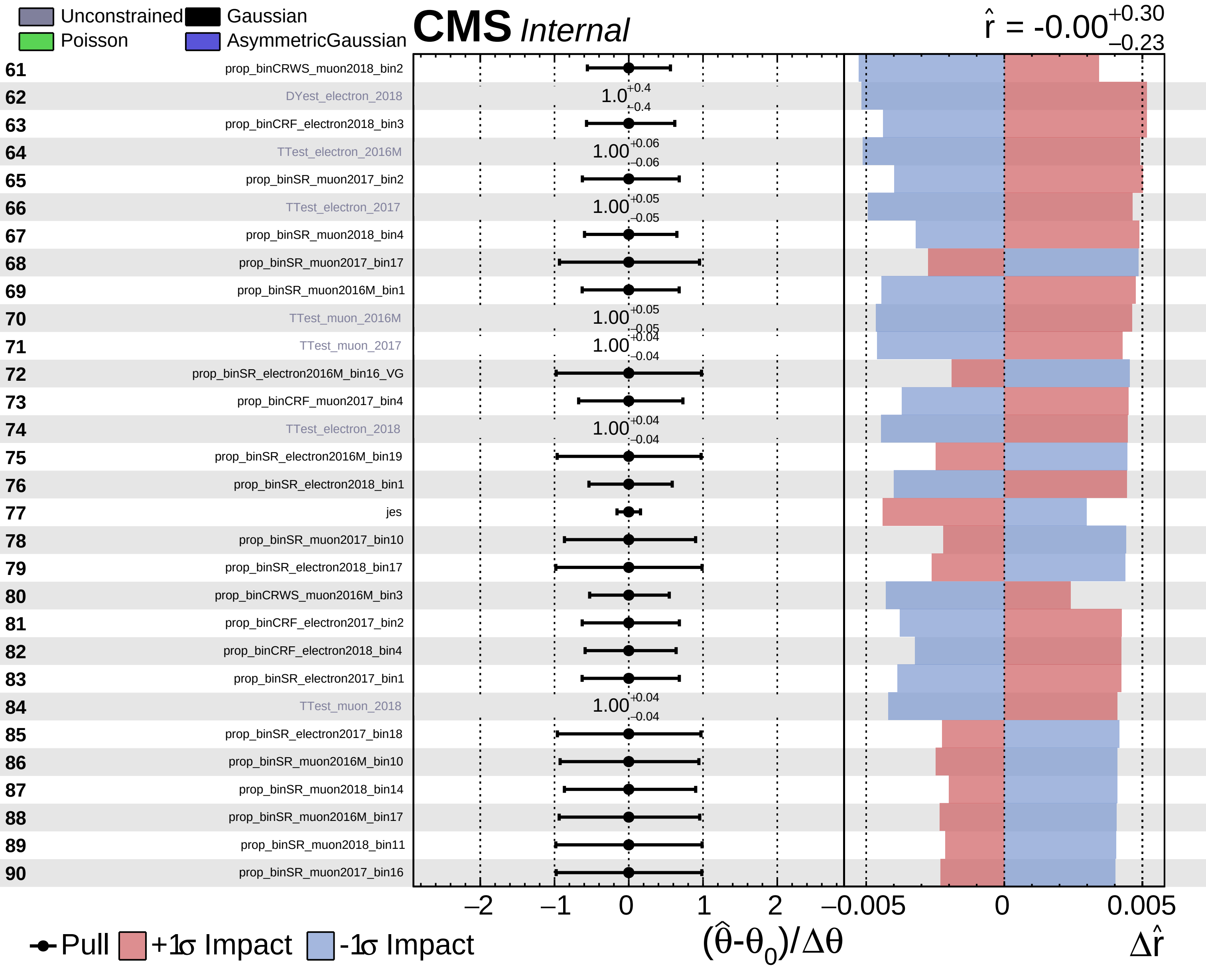
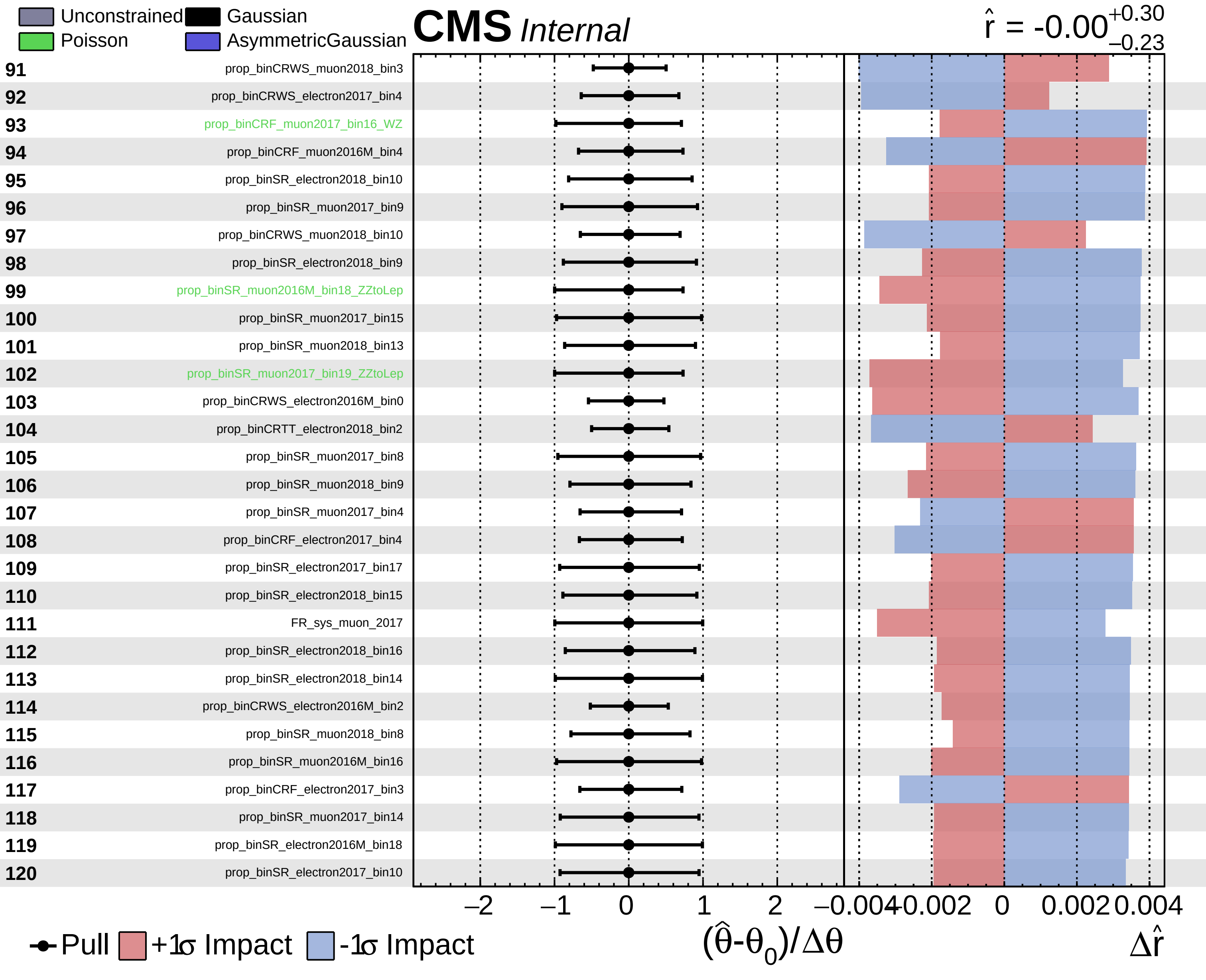

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
1 pdf_total
2 FRest_muon_2018 $1.00^{+0.08}_{-0.08}$
3 prop_binCRF_muon2018_bin0
4 FRest_muon_2017 $1.00^{+0.10}_{-0.09}$
5 prop_binSR_muon2018_bin20
6 FRest_muon_2016M $1.00^{+0.10}_{-0.09}$
7 prop_binCRF_muon2017_bin0
8 prop_binSR_muon2018_bin0
9 prop_binSR_electron2017_bin14_TVX
10 prop_binCRF_muon2016M_bin0
11 FRest_electron_2018 $1.00^{+0.08}_{-0.08}$
12 prop_binSR_electron2018_bin20
13 prop_binCRF_electron2018_bin0
14 prop_binCRF_muon2018_bin1
15 prop_binSR_muon2017_bin20
16 FRest_electron_2017 $1.00^{+0.10}_{-0.10}$
17 FRest_electron_2016M $1.00^{+0.10}_{-0.10}$
18 prop_binCRF_electron2017_bin0
19 prop_binSR_muon2017_bin0
20 prop_binCRF_electron2016M_bin0
21 prop_binSR_muon2017_bin19_WrongSign
22 prop_binCRF_muon2018_bin2
23 prop_binCRF_muon2017_bin1
24 prop_binSR_muon2016M_bin20
25 prop_binSR_muon2018_bin19
26 prop_binSR_electron2017_bin20
27 prop_binCRF_electron2018_bin1
28 prop_binCRF_muon2016M_bin1
29 prop_binCRF_muon2018_bin3
30 prop_binSR_electron2018_bin0
–2 –1 0 1 2
–0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

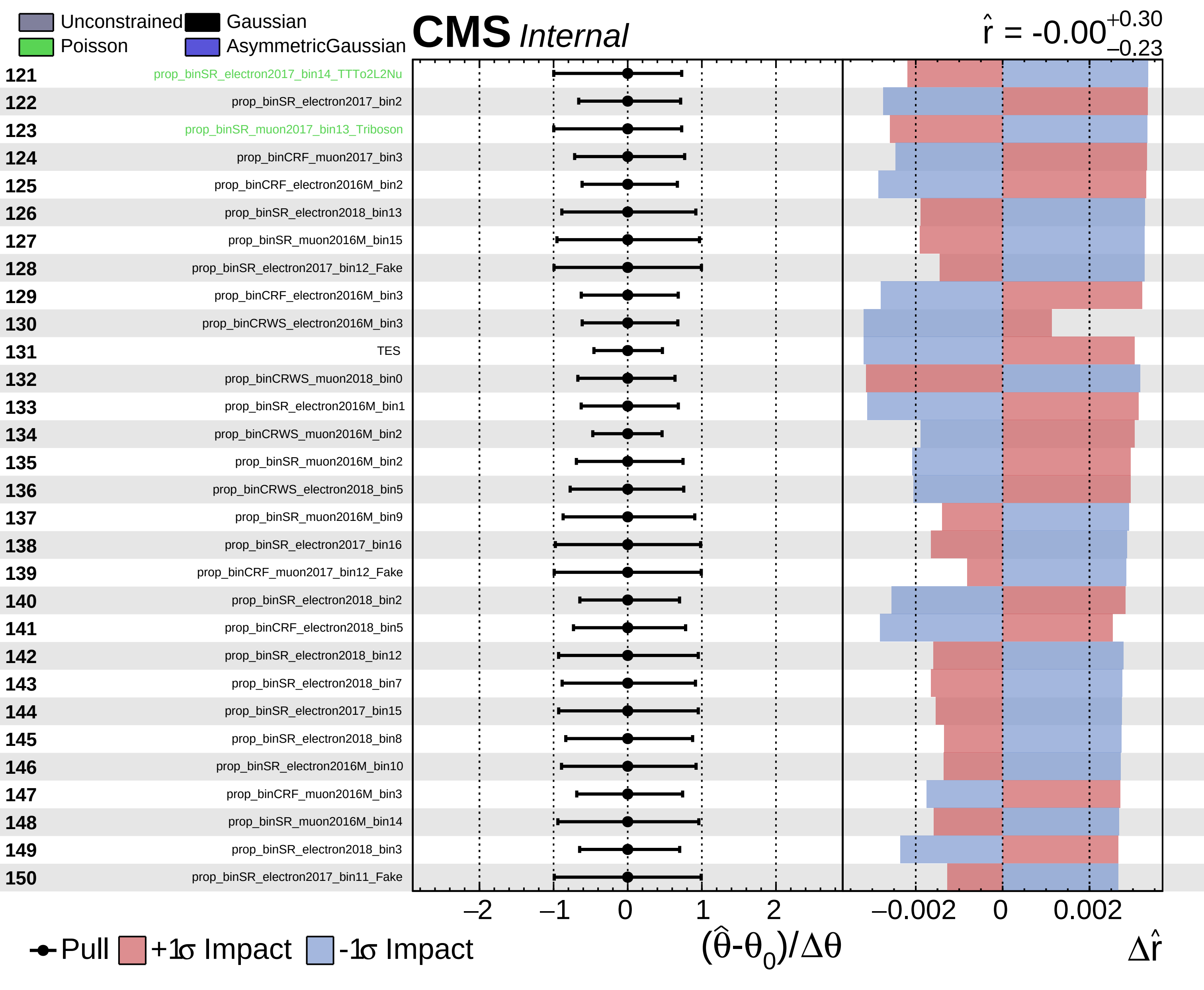

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
31 prop_binSR_muon2018_bin1
32 prop_binSR_muon2017_bin1
33 prop_binSR_muon2016M_bin0
34 prop_binSR_muon2018_bin17
35 prop_binCRF_electron2017_bin1
36 prop_binSR_electron2018_bin19
37 prop_binSR_electron2016M_bin20
38 DYest_muon_2017 $1.00^{+0.26}_{-0.26}$
39 prop_binCRF_muon2018_bin4
40 prop_binSR_muon2018_bin18
41 prop_binSR_electron2017_bin0
42 prop_binCRF_electron2018_bin2
43 prop_binCRF_electron2016M_bin1
44 prop_binSR_electron2017_bin19_TTTo2L2Nu
45 prop_binCRF_muon2016M_bin2
46 DYest_muon_2018 $1.00^{+0.20}_{-0.20}$
47 DYest_muon_2016M $1.00^{+0.32}_{-0.32}$
48 prop_binSR_muon2018_bin2
49 prop_binSR_muon2018_bin3
50 DYest_electron_2016M $1.00^{+0.34}_{-0.34}$
51 prop_binCRF_muon2017_bin2
52 prop_binSR_muon2018_bin16
53 prop_binSR_muon2016M_bin19
54 jer_2018
55 prop_binSR_muon2018_bin10
56 prop_binSR_muon2018_bin15
57 DYest_electron_2017 $1.00^{+0.30}_{-0.30}$
58 prop_binSR_electron2018_bin18
59 prop_binSR_muon2017_bin18
60 prop_binSR_electron2016M_bin0
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

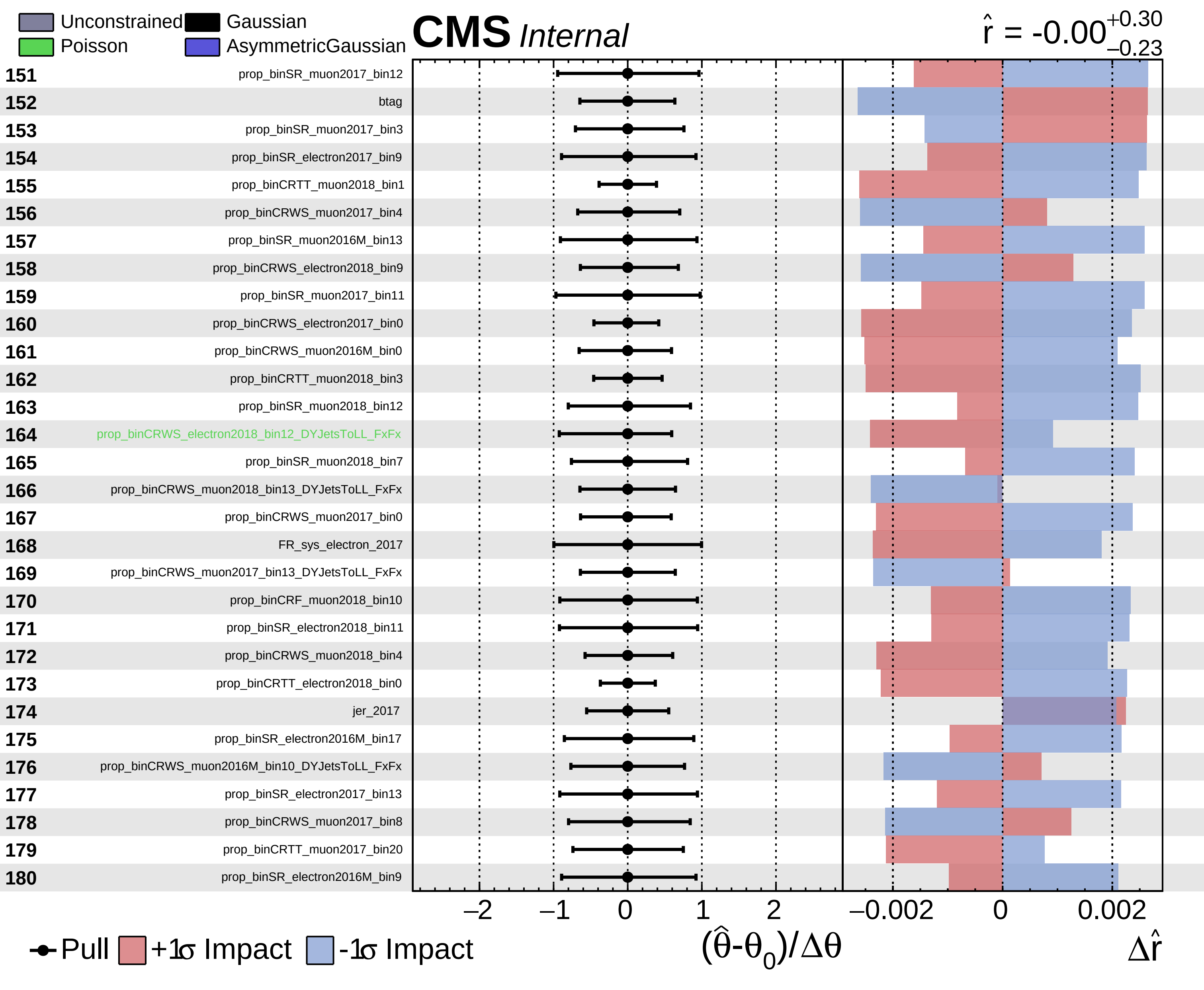


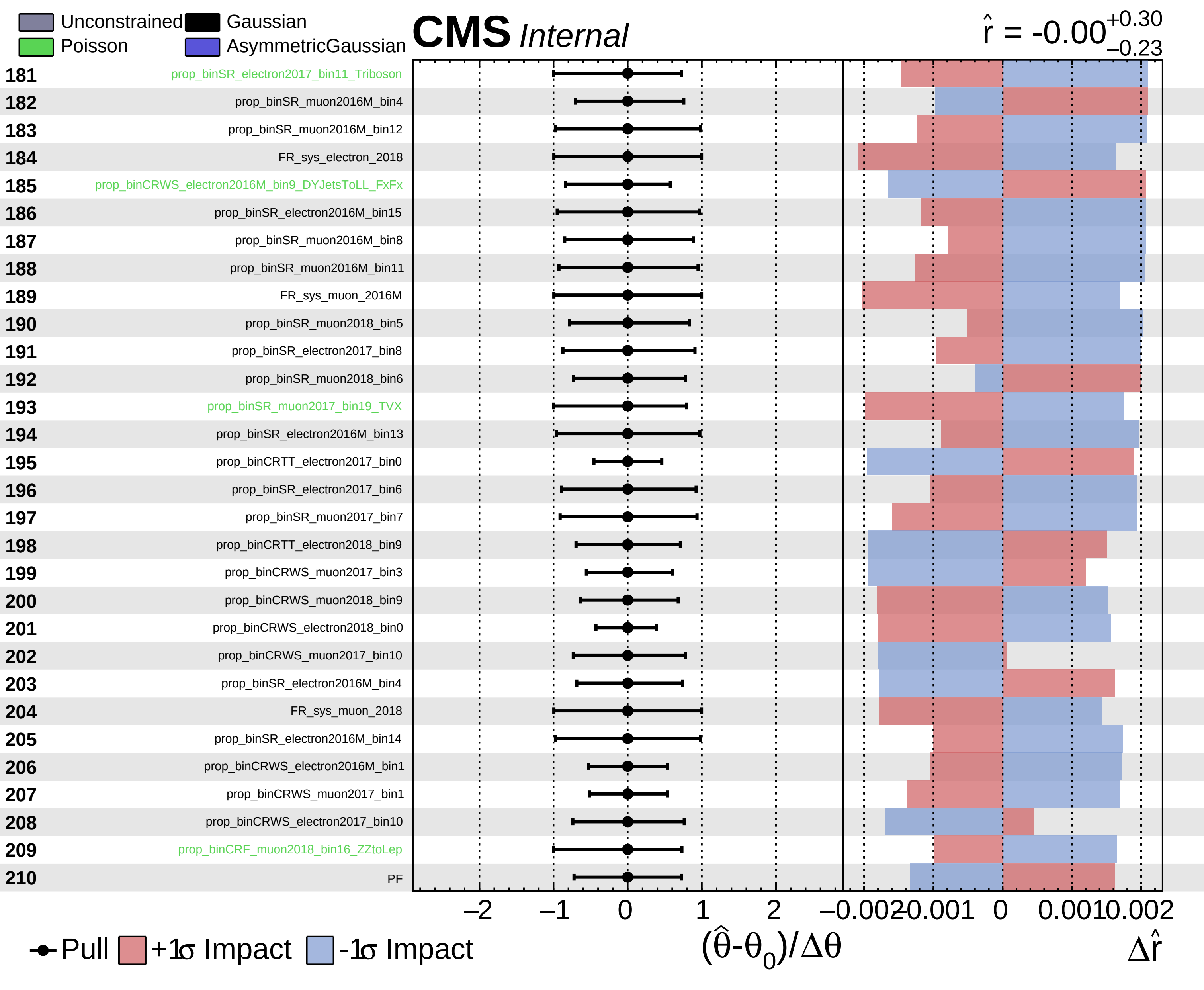
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
61 prop_binCRWS_muon2018_bin2
62 DYest_electron_2018 $1.0^{+0.4}_{-0.4}$
63 prop_binCRF_electron2018_bin3
64 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
65 prop_binSR_muon2017_bin2
66 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
67 prop_binSR_muon2018_bin4
68 prop_binSR_muon2017_bin17
69 prop_binSR_muon2016M_bin1
70 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
71 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
72 prop_binSR_electron2016M_bin16_VG
73 prop_binCRF_muon2017_bin4
74 TTest_electron_2018 $1.00^{+0.04}_{-0.04}$
75 prop_binSR_electron2016M_bin19
76 prop_binSR_electron2018_bin1
77 jes
78 prop_binSR_muon2017_bin10
79 prop_binSR_electron2018_bin17
80 prop_binCRWS_muon2016M_bin3
81 prop_binCRF_electron2017_bin2
82 prop_binCRF_electron2018_bin4
83 prop_binSR_electron2017_bin1
84 TTest_muon_2018 $1.00^{+0.04}_{-0.04}$
85 prop_binSR_electron2017_bin18
86 prop_binSR_muon2016M_bin10
87 prop_binSR_muon2018_bin14
88 prop_binSR_muon2016M_bin17
89 prop_binSR_muon2018_bin11
90 prop_binSR_muon2017_bin16
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



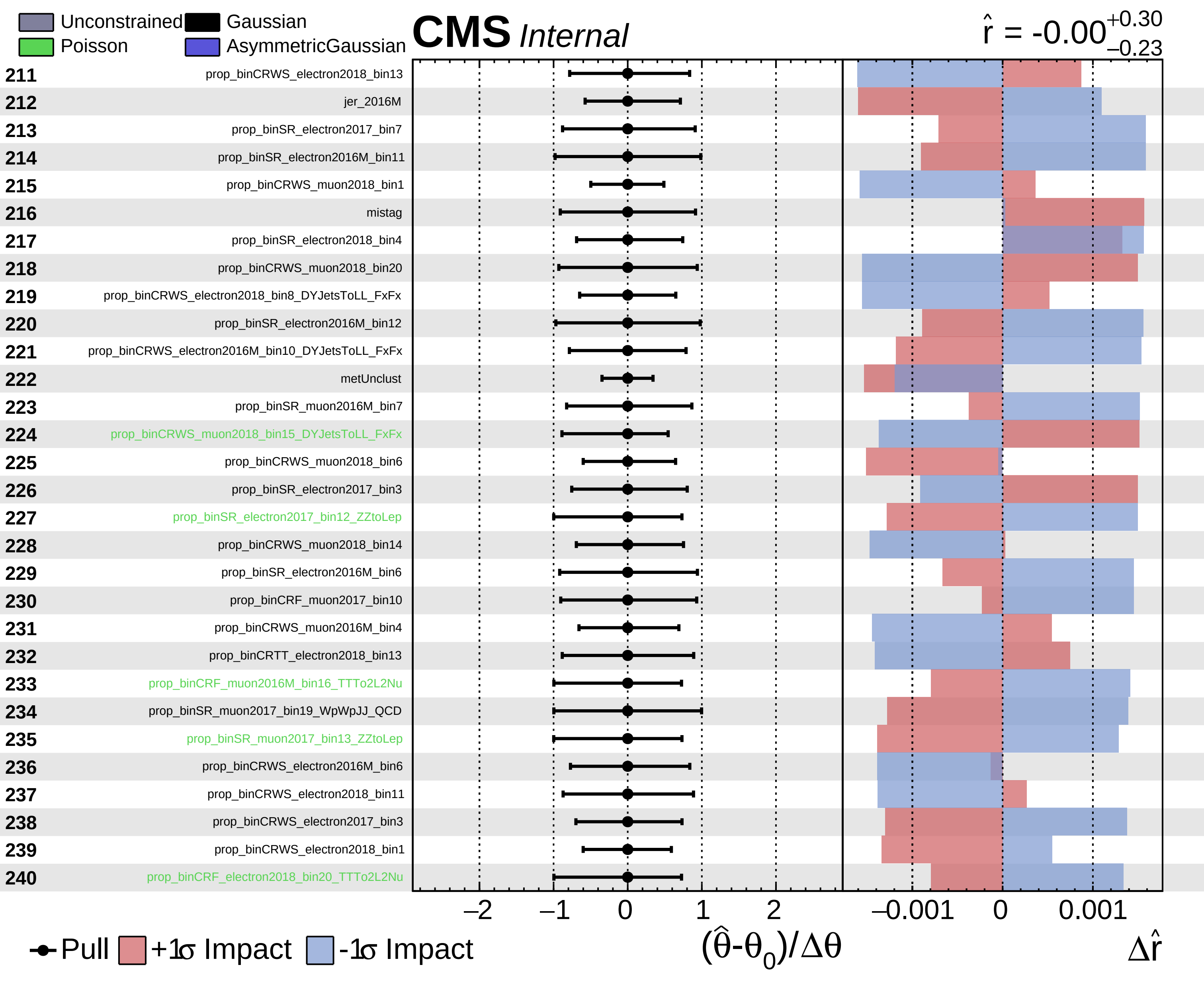
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
91 prop_binCRWS_muon2018_bin3
92 prop_binCRWS_electron2017_bin4
93 prop_binCRF_muon2017_bin16_WZ
94 prop_binCRF_muon2016M_bin4
95 prop_binSR_electron2018_bin10
96 prop_binSR_muon2017_bin9
97 prop_binCRWS_muon2018_bin10
98 prop_binSR_electron2018_bin9
99 prop_binSR_muon2016M_bin18_ZZtoLep
100 prop_binSR_muon2017_bin15
101 prop_binSR_muon2018_bin13
102 prop_binSR_muon2017_bin19_ZZtoLep
103 prop_binCRWS_electron2016M_bin0
104 prop_binCRTT_electron2018_bin2
105 prop_binSR_muon2017_bin8
106 prop_binSR_muon2018_bin9
107 prop_binSR_muon2017_bin4
108 prop_binCRF_electron2017_bin4
109 prop_binSR_electron2017_bin17
110 prop_binSR_electron2018_bin15
111 FR_sys_muon_2017
112 prop_binSR_electron2018_bin16
113 prop_binSR_electron2018_bin14
114 prop_binCRWS_electron2016M_bin2
115 prop_binSR_muon2018_bin8
116 prop_binSR_muon2016M_bin16
117 prop_binCRF_electron2017_bin3
118 prop_binSR_muon2017_bin14
119 prop_binSR_electron2016M_bin18
120 prop_binSR_electron2017_bin10
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
121 prop_binSR_electron2017_bin14_TTTo2L2Nu
122 prop_binSR_electron2017_bin2
123 prop_binSR_muon2017_bin13_Triboson
124 prop_binCRF_muon2017_bin3
125 prop_binCRF_electron2016M_bin2
126 prop_binSR_electron2018_bin13
127 prop_binSR_muon2016M_bin15
128 prop_binSR_electron2017_bin12_Fake
129 prop_binCRF_electron2016M_bin3
130 prop_binCRWS_electron2016M_bin3
131 TES
132 prop_binCRWS_muon2018_bin0
133 prop_binSR_electron2016M_bin1
134 prop_binCRWS_muon2016M_bin2
135 prop_binSR_muon2016M_bin2
136 prop_binCRWS_electron2018_bin5
137 prop_binSR_muon2016M_bin9
138 prop_binSR_electron2017_bin16
139 prop_binCRF_muon2017_bin12_Fake
140 prop_binSR_electron2018_bin2
141 prop_binCRF_electron2018_bin5
142 prop_binSR_electron2018_bin12
143 prop_binSR_electron2018_bin7
144 prop_binSR_electron2017_bin15
145 prop_binSR_electron2018_bin8
146 prop_binSR_electron2016M_bin10
147 prop_binCRF_muon2016M_bin3
148 prop_binSR_muon2016M_bin14
149 prop_binSR_electron2018_bin3
150 prop_binSR_electron2017_bin11_Fake
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

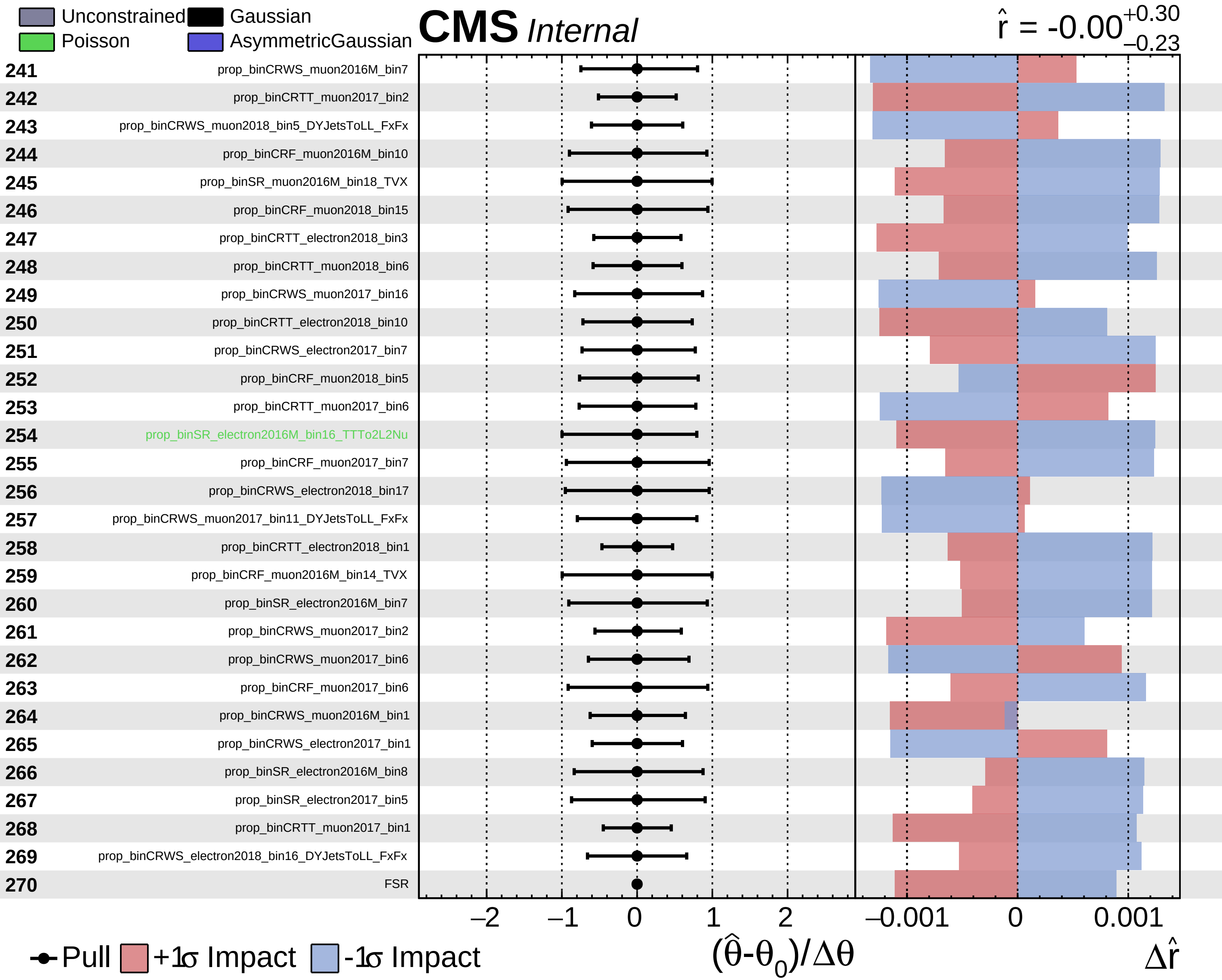

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.30 −0.23
151 prop_binSR_muon2017_bin12
152 btag
153 prop_binSR_muon2017_bin3
154 prop_binSR_electron2017_bin9
155 prop_binCRTT_muon2018_bin1
156 prop_binCRWS_muon2017_bin4
157 prop_binSR_muon2016M_bin13
158 prop_binCRWS_electron2018_bin9
159 prop_binSR_muon2017_bin11
160 prop_binCRWS_electron2017_bin0
161 prop_binCRWS_muon2016M_bin0
162 prop_binCRTT_muon2018_bin3
163 prop_binSR_muon2018_bin12
164 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
165 prop_binSR_muon2018_bin7
166 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
167 prop_binCRWS_muon2017_bin0
168 FR_sys_electron_2017
169 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
170 prop_binCRF_muon2018_bin10
171 prop_binSR_electron2018_bin11
172 prop_binCRWS_muon2018_bin4
173 prop_binCRTT_electron2018_bin0
174 jer_2017
175 prop_binSR_electron2016M_bin17
176 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
177 prop_binSR_electron2017_bin13
178 prop_binCRWS_muon2017_bin8
179 prop_binCRTT_muon2017_bin20
180 prop_binSR_electron2016M_bin9
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



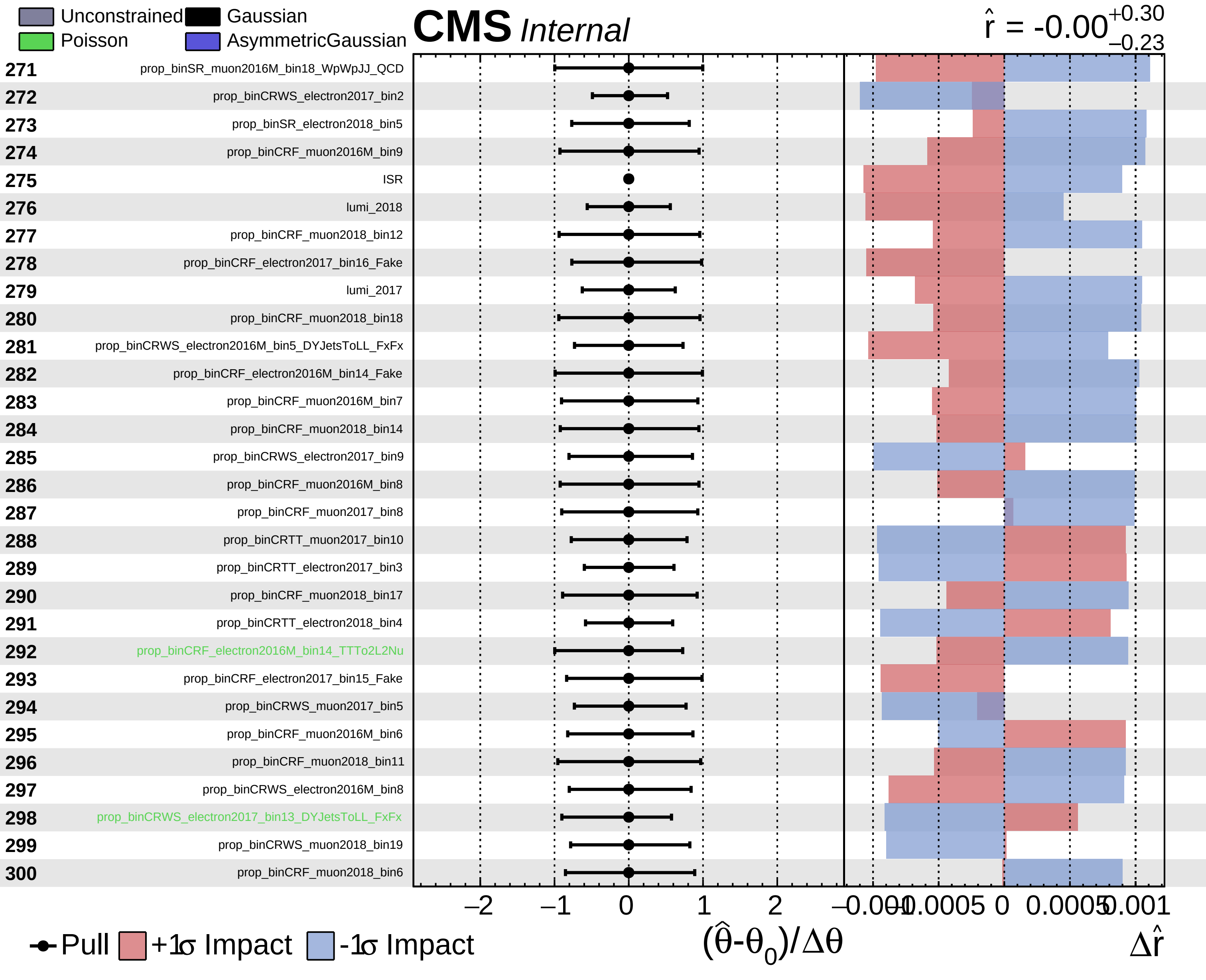
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
181 prop_binSR_electron2017_bin11_Triboson
182 prop_binSR_muon2016M_bin4
183 prop_binSR_muon2016M_bin12
184 FR_sys_electron_2018
185 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
186 prop_binSR_electron2016M_bin15
187 prop_binSR_muon2016M_bin8
188 prop_binSR_muon2016M_bin11
189 FR_sys_muon_2016M
190 prop_binSR_muon2018_bin5
191 prop_binSR_electron2017_bin8
192 prop_binSR_muon2018_bin6
193 prop_binSR_muon2017_bin19_TVX
194 prop_binSR_electron2016M_bin13
195 prop_binCRTT_electron2017_bin0
196 prop_binSR_electron2017_bin6
197 prop_binSR_muon2017_bin7
198 prop_binCRTT_electron2018_bin9
199 prop_binCRWS_muon2017_bin3
200 prop_binCRWS_muon2018_bin9
201 prop_binCRWS_electron2018_bin0
202 prop_binCRWS_muon2017_bin10
203 prop_binSR_electron2016M_bin4
204 FR_sys_muon_2018
205 prop_binSR_electron2016M_bin14
206 prop_binCRWS_electron2016M_bin1
207 prop_binCRWS_muon2017_bin1
208 prop_binCRWS_electron2017_bin10
209 prop_binCRF_muon2018_bin16_ZZtoLep
210 PF
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

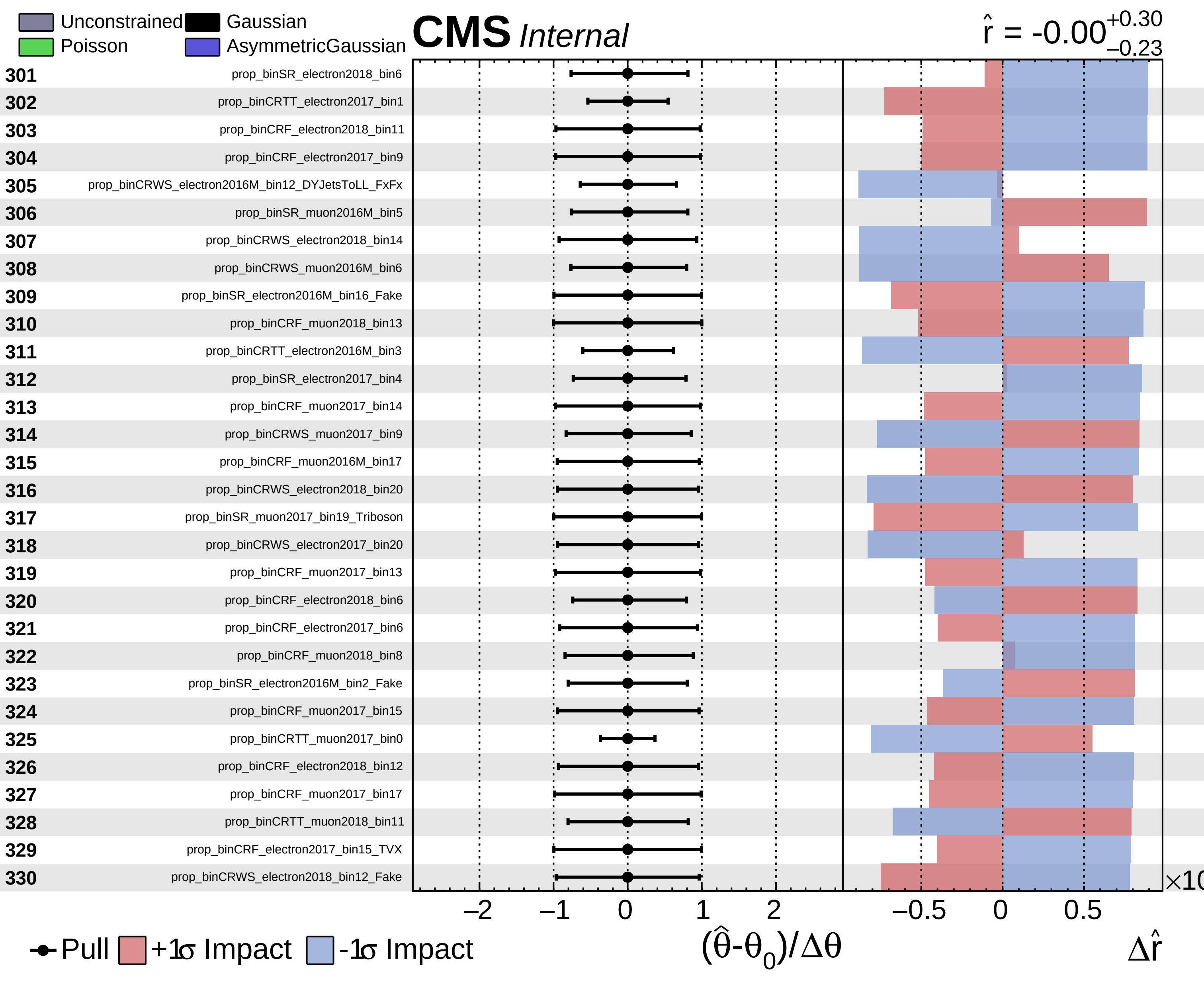


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
211 prop_binCRWS_electron2018_bin13
212 jer_2016M
213 prop_binSR_electron2017_bin7
214 prop_binSR_electron2016M_bin11
215 prop_binCRWS_muon2018_bin1
216 mistag
217 prop_binSR_electron2018_bin4
218 prop_binCRWS_muon2018_bin20
219 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
220 prop_binSR_electron2016M_bin12
221 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
222 metUnclust
223 prop_binSR_muon2016M_bin7
224 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
225 prop_binCRWS_muon2018_bin6
226 prop_binSR_electron2017_bin3
227 prop_binSR_electron2017_bin12_ZZtoLep
228 prop_binCRWS_muon2018_bin14
229 prop_binSR_electron2016M_bin6
230 prop_binCRF_muon2017_bin10
231 prop_binCRWS_muon2016M_bin4
232 prop_binCRTT_electron2018_bin13
233 prop_binCRF_muon2016M_bin16_TTTo2L2Nu
234 prop_binSR_muon2017_bin19_WpWpJJ_QCD
235 prop_binSR_muon2017_bin13_ZZtoLep
236 prop_binCRWS_electron2016M_bin6
237 prop_binCRWS_electron2018_bin11
238 prop_binCRWS_electron2017_bin3
239 prop_binCRWS_electron2018_bin1
240 prop_binCRF_electron2018_bin20_TTTo2L2Nu
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

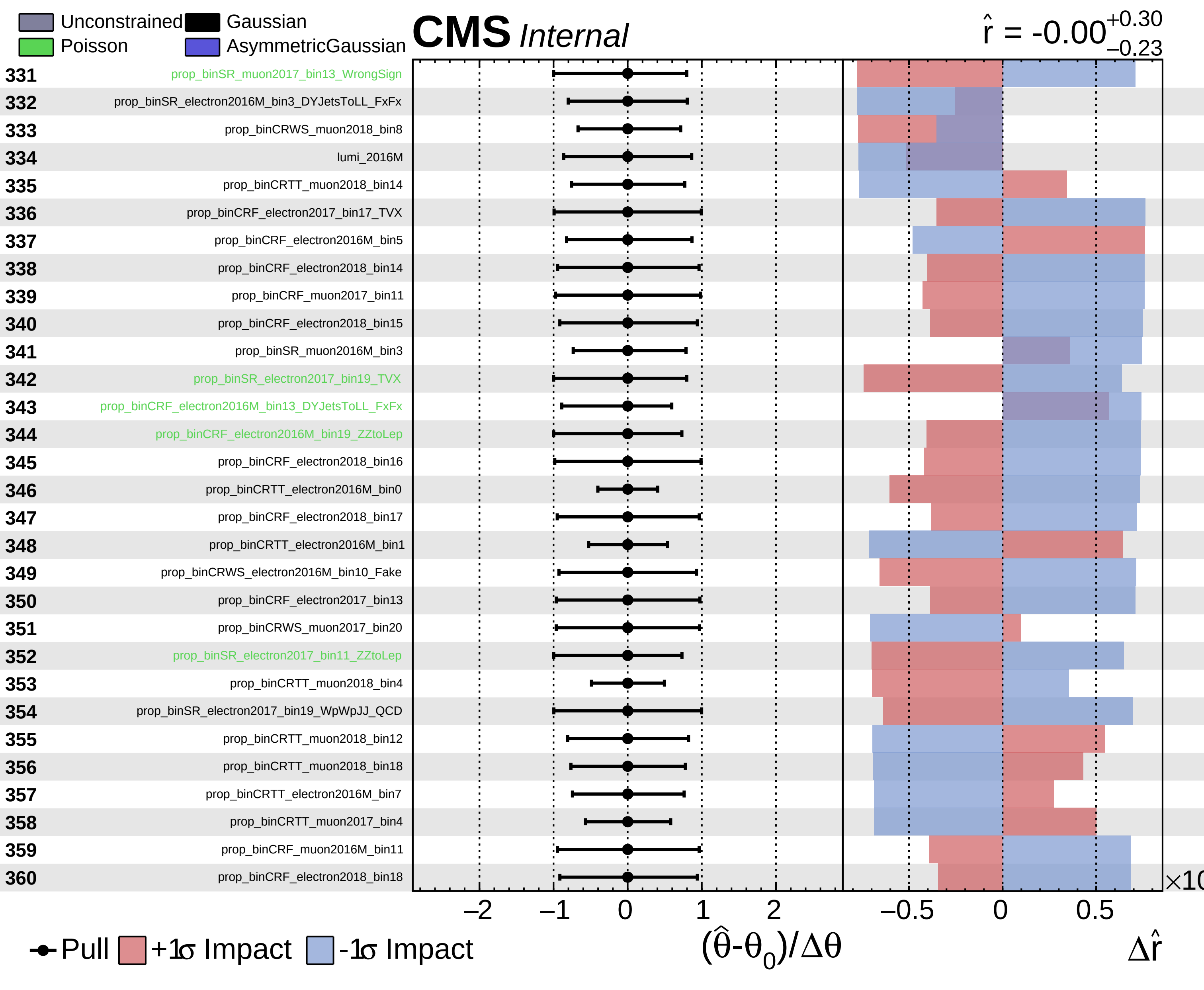


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
241 prop_binCRWS_muon2016M_bin7
242 prop_binCRTT_muon2017_bin2
243 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
244 prop_binCRF_muon2016M_bin10
245 prop_binSR_muon2016M_bin18_TVX
246 prop_binCRF_muon2018_bin15
247 prop_binCRTT_electron2018_bin3
248 prop_binCRTT_muon2018_bin6
249 prop_binCRWS_muon2017_bin16
250 prop_binCRTT_electron2018_bin10
251 prop_binCRWS_electron2017_bin7
252 prop_binCRF_muon2018_bin5
253 prop_binCRTT_muon2017_bin6
254 prop_binSR_electron2016M_bin16_TTTo2L2Nu
255 prop_binCRF_muon2017_bin7
256 prop_binCRWS_electron2018_bin17
257 prop_binCRWS_muon2017_bin11_DYJetsToLL_FxFx
258 prop_binCRTT_electron2018_bin1
259 prop_binCRF_muon2016M_bin14_TVX
260 prop_binSR_electron2016M_bin7
261 prop_binCRWS_muon2017_bin2
262 prop_binCRWS_muon2017_bin6
263 prop_binCRF_muon2017_bin6
264 prop_binCRWS_muon2016M_bin1
265 prop_binCRWS_electron2017_bin1
266 prop_binSR_electron2016M_bin8
267 prop_binSR_electron2017_bin5
268 prop_binCRTT_muon2017_bin1
269 prop_binCRWS_electron2018_bin16_DYJetsToLL_FxFx
270 FSR
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

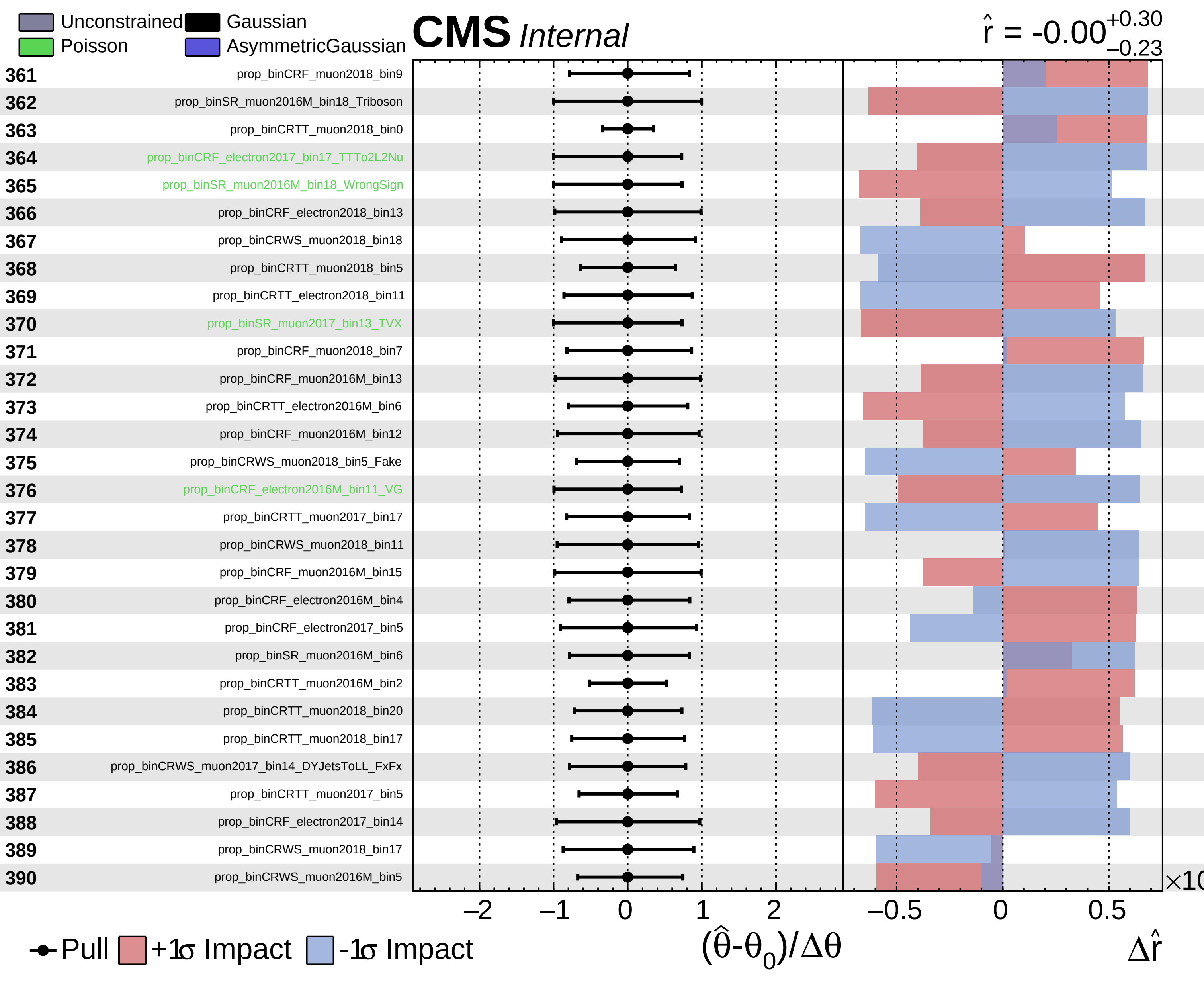

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
271 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
272 prop_binCRWS_electron2017_bin2
273 prop_binSR_electron2018_bin5
274 prop_binCRF_muon2016M_bin9
275 ISR
276 lumi_2018
277 prop_binCRF_muon2018_bin12
278 prop_binCRF_electron2017_bin16_Fake
279 lumi_2017
280 prop_binCRF_muon2018_bin18
281 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
282 prop_binCRF_electron2016M_bin14_Fake
283 prop_binCRF_muon2016M_bin7
284 prop_binCRF_muon2018_bin14
285 prop_binCRWS_electron2017_bin9
286 prop_binCRF_muon2016M_bin8
287 prop_binCRF_muon2017_bin8
288 prop_binCRTT_muon2017_bin10
289 prop_binCRTT_electron2017_bin3
290 prop_binCRF_muon2018_bin17
291 prop_binCRTT_electron2018_bin4
292 prop_binCRF_electron2016M_bin14_TTTo2L2Nu
293 prop_binCRF_electron2017_bin15_Fake
294 prop_binCRWS_muon2017_bin5
295 prop_binCRF_muon2016M_bin6
296 prop_binCRF_muon2018_bin11
297 prop_binCRWS_electron2016M_bin8
298 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
299 prop_binCRWS_muon2018_bin19
300 prop_binCRF_muon2018_bin6
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



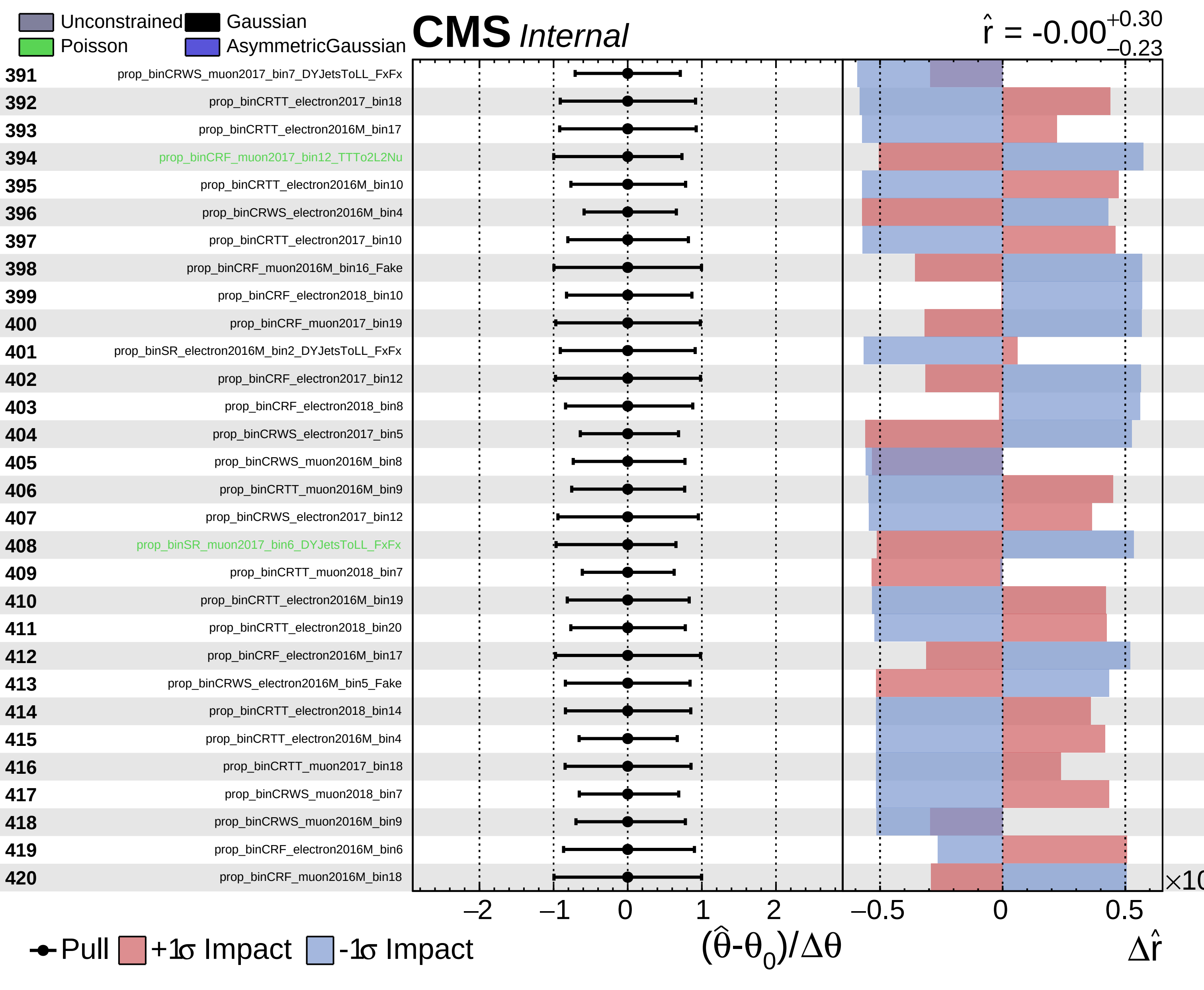
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
301 prop_binSR_electron2018_bin6
302 prop_binCRTT_electron2017_bin1
303 prop_binCRF_electron2018_bin11
304 prop_binCRF_electron2017_bin9
305 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
306 prop_binSR_muon2016M_bin5
307 prop_binCRWS_electron2018_bin14
308 prop_binCRWS_muon2016M_bin6
309 prop_binSR_electron2016M_bin16_Fake
310 prop_binCRF_muon2018_bin13
311 prop_binCRTT_electron2016M_bin3
312 prop_binSR_electron2017_bin4
313 prop_binCRF_muon2017_bin14
314 prop_binCRWS_muon2017_bin9
315 prop_binCRF_muon2016M_bin17
316 prop_binCRWS_electron2018_bin20
317 prop_binSR_muon2017_bin19_Triboson
318 prop_binCRWS_electron2017_bin20
319 prop_binCRF_muon2017_bin13
320 prop_binCRF_electron2018_bin6
321 prop_binCRF_electron2017_bin6
322 prop_binCRF_muon2018_bin8
323 prop_binSR_electron2016M_bin2_Fake
324 prop_binCRF_muon2017_bin15
325 prop_binCRTT_muon2017_bin0
326 prop_binCRF_electron2018_bin12
327 prop_binCRF_muon2017_bin17
328 prop_binCRTT_muon2018_bin11
329 prop_binCRF_electron2017_bin15_TVX
330 prop_binCRWS_electron2018_bin12_Fake
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

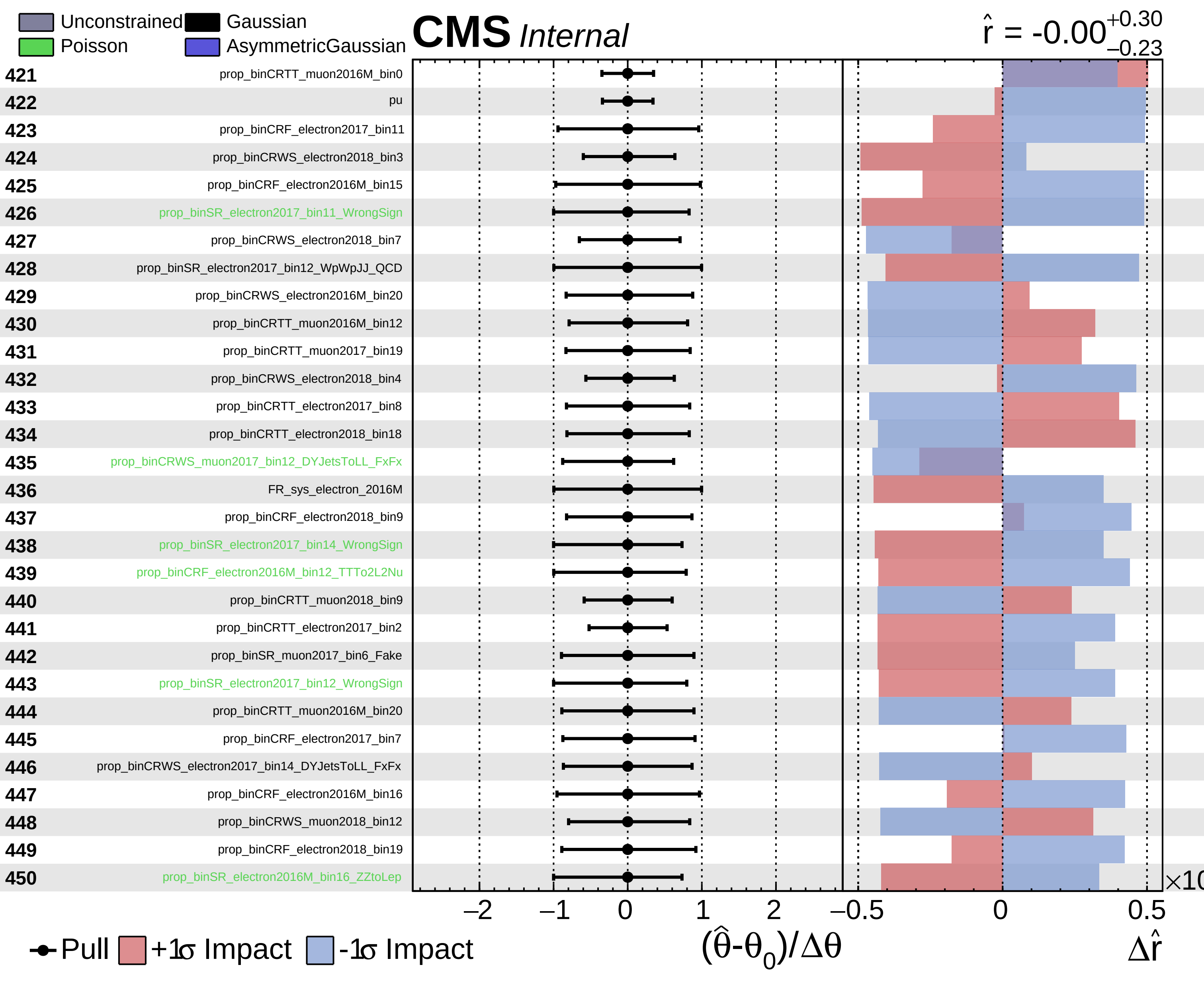


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
331 prop_binSR_muon2017_bin13_WrongSign
332 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
333 prop_binCRWS_muon2018_bin8
334 lumi_2016M
335 prop_binCRTT_muon2018_bin14
336 prop_binCRF_electron2017_bin17_TVX
337 prop_binCRF_electron2016M_bin5
338 prop_binCRF_electron2018_bin14
339 prop_binCRF_muon2017_bin11
340 prop_binCRF_electron2018_bin15
341 prop_binSR_muon2016M_bin3
342 prop_binSR_electron2017_bin19_TVX
343 prop_binCRF_electron2016M_bin13_DYJetsToLL_FxFx
344 prop_binCRF_electron2016M_bin19_ZZtoLep
345 prop_binCRF_electron2018_bin16
346 prop_binCRTT_electron2016M_bin0
347 prop_binCRF_electron2018_bin17
348 prop_binCRTT_electron2016M_bin1
349 prop_binCRWS_electron2016M_bin10_Fake
350 prop_binCRF_electron2017_bin13
351 prop_binCRWS_muon2017_bin20
352 prop_binSR_electron2017_bin11_ZZtoLep
353 prop_binCRTT_muon2018_bin4
354 prop_binSR_electron2017_bin19_WpWpJJ_QCD
355 prop_binCRTT_muon2018_bin12
356 prop_binCRTT_muon2018_bin18
357 prop_binCRTT_electron2016M_bin7
358 prop_binCRTT_muon2017_bin4
359 prop_binCRF_muon2016M_bin11
360 prop_binCRF_electron2018_bin18
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

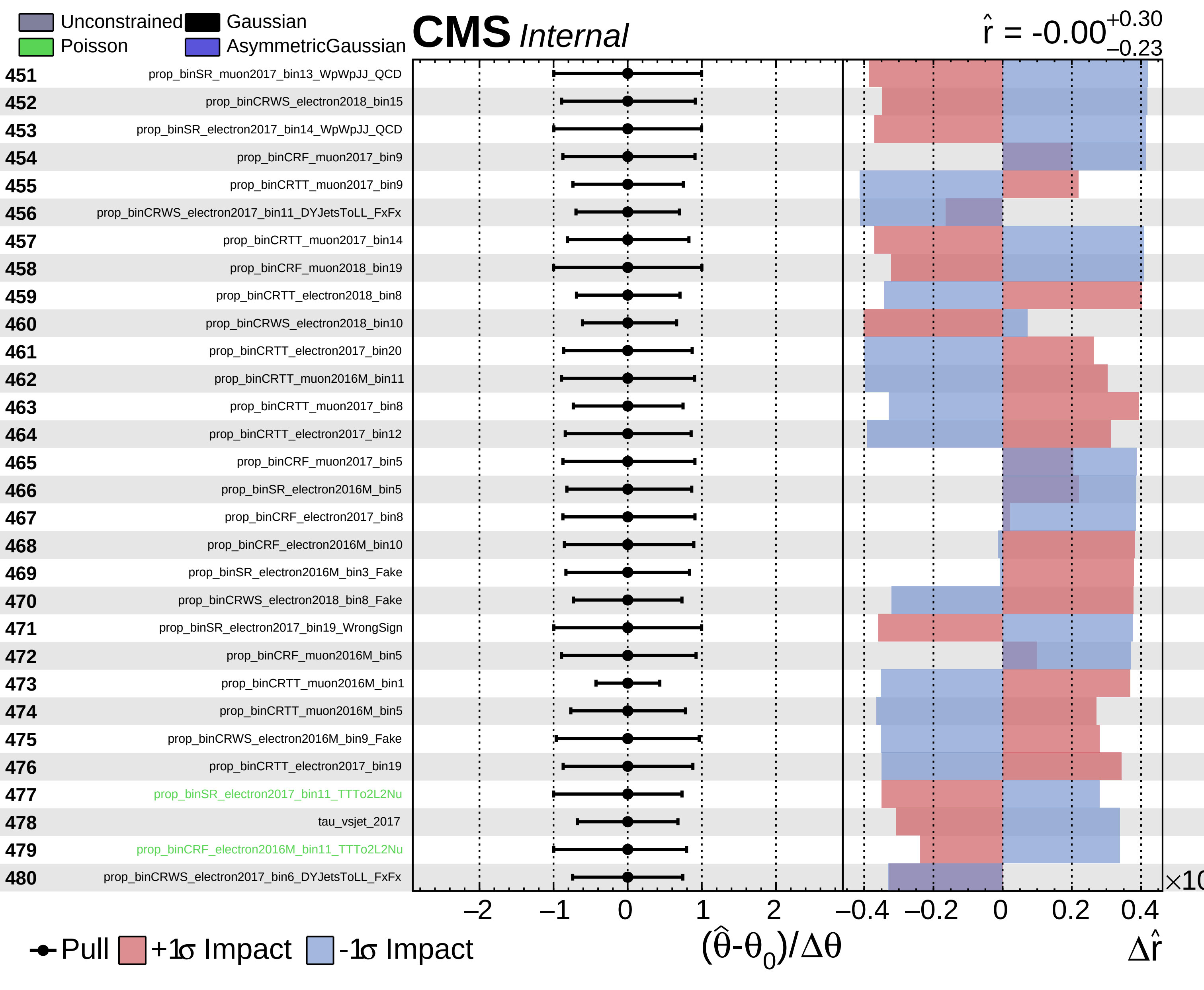


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
361 prop_binCRF_muon2018_bin9
362 prop_binSR_muon2016M_bin18_Triboson
363 prop_binCRTT_muon2018_bin0
364 prop_binCRF_electron2017_bin17_TTTo2L2Nu
365 prop_binSR_muon2016M_bin18_WrongSign
366 prop_binCRF_electron2018_bin13
367 prop_binCRWS_muon2018_bin18
368 prop_binCRTT_muon2018_bin5
369 prop_binCRTT_electron2018_bin11
370 prop_binSR_muon2017_bin13_TVX
371 prop_binCRF_muon2018_bin7
372 prop_binCRF_muon2016M_bin13
373 prop_binCRTT_electron2016M_bin6
374 prop_binCRF_muon2016M_bin12
375 prop_binCRWS_muon2018_bin5_Fake
376 prop_binCRF_electron2016M_bin11_VG
377 prop_binCRTT_muon2017_bin17
378 prop_binCRWS_muon2018_bin11
379 prop_binCRF_muon2016M_bin15
380 prop_binCRF_electron2016M_bin4
381 prop_binCRF_electron2017_bin5
382 prop_binSR_muon2016M_bin6
383 prop_binCRTT_muon2016M_bin2
384 prop_binCRTT_muon2018_bin20
385 prop_binCRTT_muon2018_bin17
386 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
387 prop_binCRTT_muon2017_bin5
388 prop_binCRF_electron2017_bin14
389 prop_binCRWS_muon2018_bin17
390 prop_binCRWS_muon2016M_bin5
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

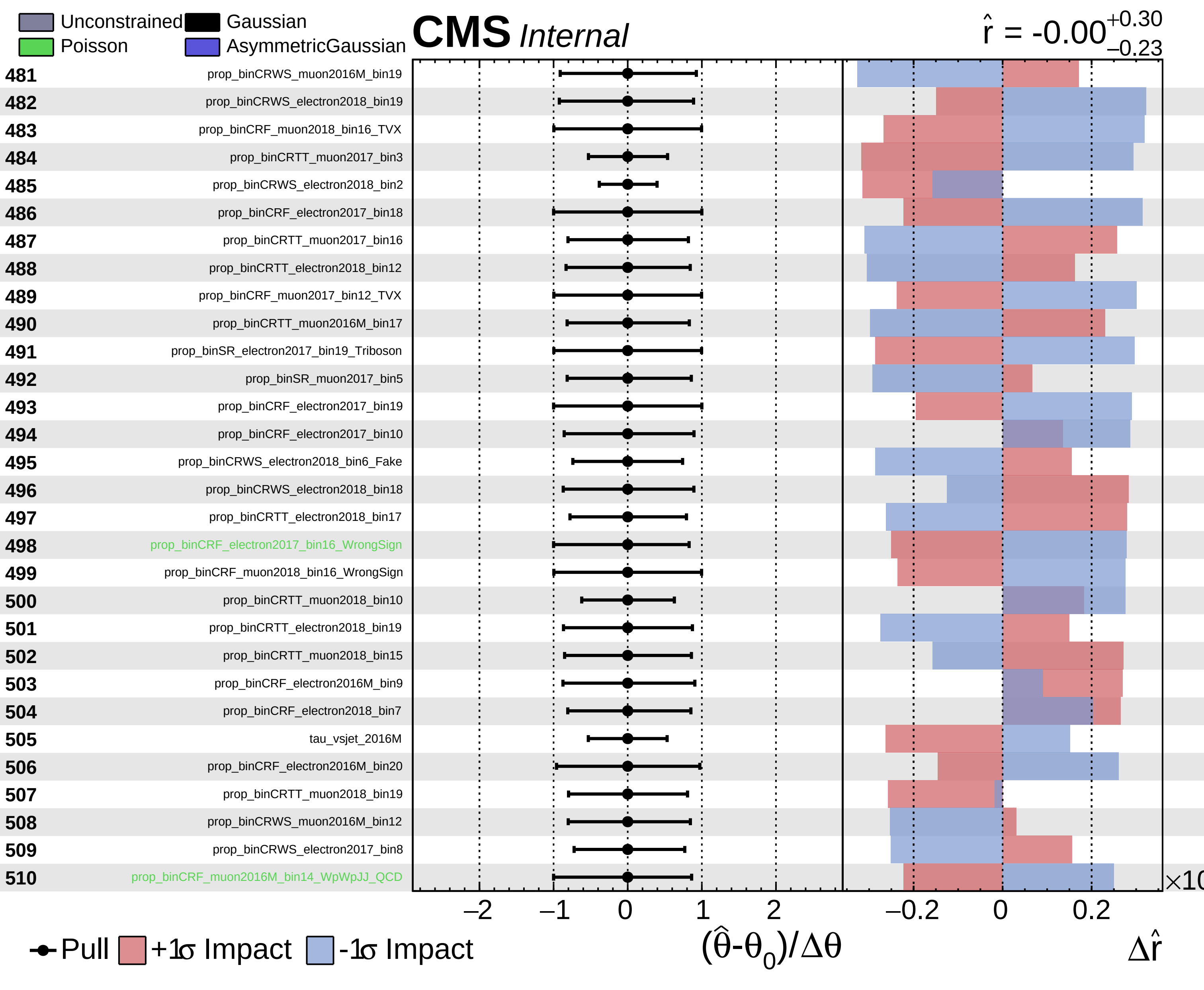

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
391 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
392 prop_binCRTT_electron2017_bin18
393 prop_binCRTT_electron2016M_bin17
394 prop_binCRF_muon2017_bin12_TTTo2L2Nu
395 prop_binCRTT_electron2016M_bin10
396 prop_binCRWS_electron2016M_bin4
397 prop_binCRTT_electron2017_bin10
398 prop_binCRF_muon2016M_bin16_Fake
399 prop_binCRF_electron2018_bin10
400 prop_binCRF_muon2017_bin19
401 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
402 prop_binCRF_electron2017_bin12
403 prop_binCRF_electron2018_bin8
404 prop_binCRWS_electron2017_bin5
405 prop_binCRWS_muon2016M_bin8
406 prop_binCRTT_muon2016M_bin9
407 prop_binCRWS_electron2017_bin12
408 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
409 prop_binCRTT_muon2018_bin7
410 prop_binCRTT_electron2016M_bin19
411 prop_binCRTT_electron2018_bin20
412 prop_binCRF_electron2016M_bin17
413 prop_binCRWS_electron2016M_bin5_Fake
414 prop_binCRTT_electron2018_bin14
415 prop_binCRTT_electron2016M_bin4
416 prop_binCRTT_muon2017_bin18
417 prop_binCRWS_muon2018_bin7
418 prop_binCRWS_muon2016M_bin9
419 prop_binCRF_electron2016M_bin6
420 prop_binCRF_muon2016M_bin18
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

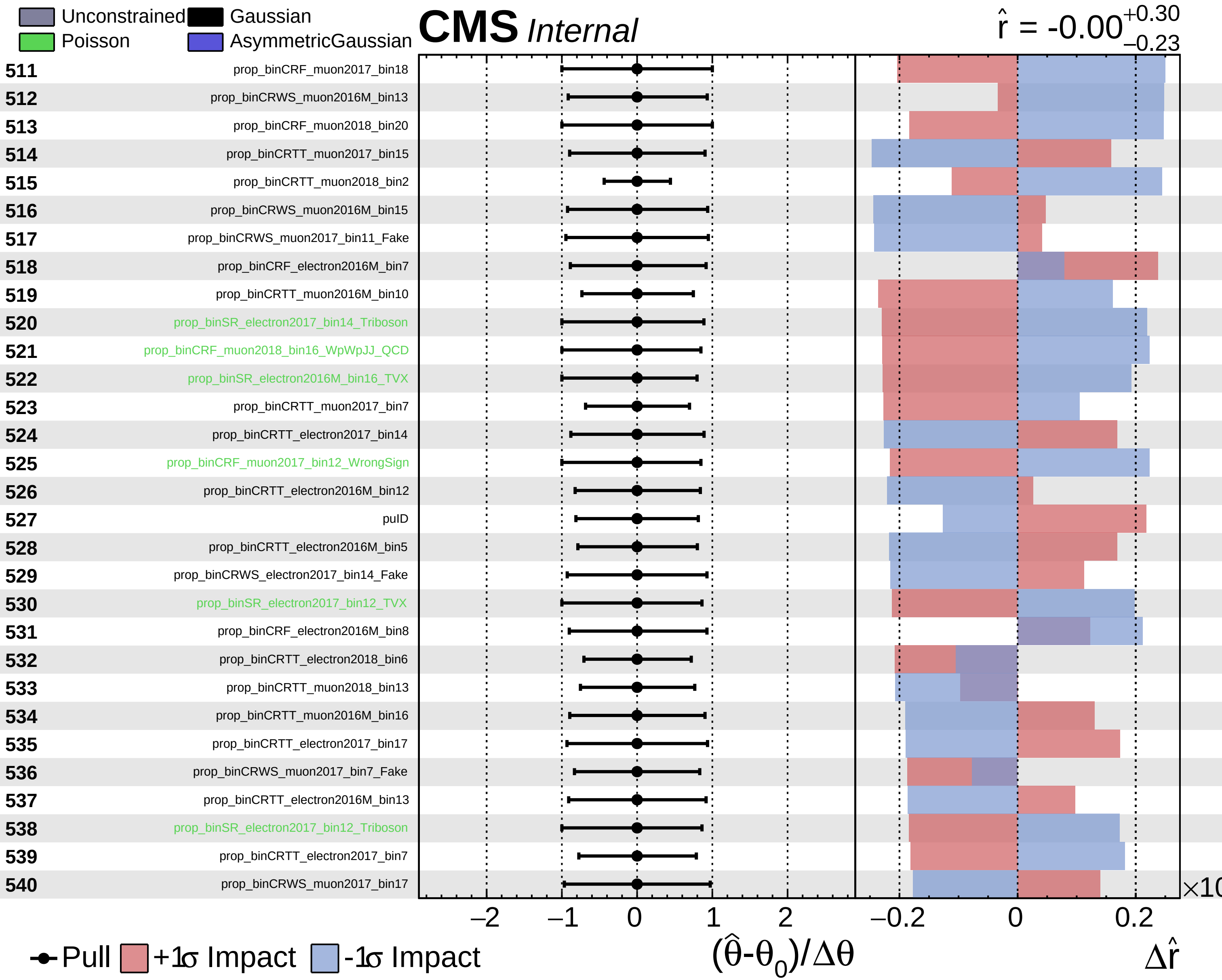


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
421 prop_binCRTT_muon2016M_bin0
422 pu
423 prop_binCRF_electron2017_bin11
424 prop_binCRWS_electron2018_bin3
425 prop_binCRF_electron2016M_bin15
426 prop_binSR_electron2017_bin11_WrongSign
427 prop_binCRWS_electron2018_bin7
428 prop_binSR_electron2017_bin12_WpWpJJ_QCD
429 prop_binCRWS_electron2016M_bin20
430 prop_binCRTT_muon2016M_bin12
431 prop_binCRTT_muon2017_bin19
432 prop_binCRWS_electron2018_bin4
433 prop_binCRTT_electron2017_bin8
434 prop_binCRTT_electron2018_bin18
435 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
436 FR_sys_electron_2016M
437 prop_binCRF_electron2018_bin9
438 prop_binSR_electron2017_bin14_WrongSign
439 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
440 prop_binCRTT_muon2018_bin9
441 prop_binCRTT_electron2017_bin2
442 prop_binSR_muon2017_bin6_Fake
443 prop_binSR_electron2017_bin12_WrongSign
444 prop_binCRTT_muon2016M_bin20
445 prop_binCRF_electron2017_bin7
446 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
447 prop_binCRF_electron2016M_bin16
448 prop_binCRWS_muon2018_bin12
449 prop_binCRF_electron2018_bin19
450 prop_binSR_electron2016M_bin16_ZZtoLep
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

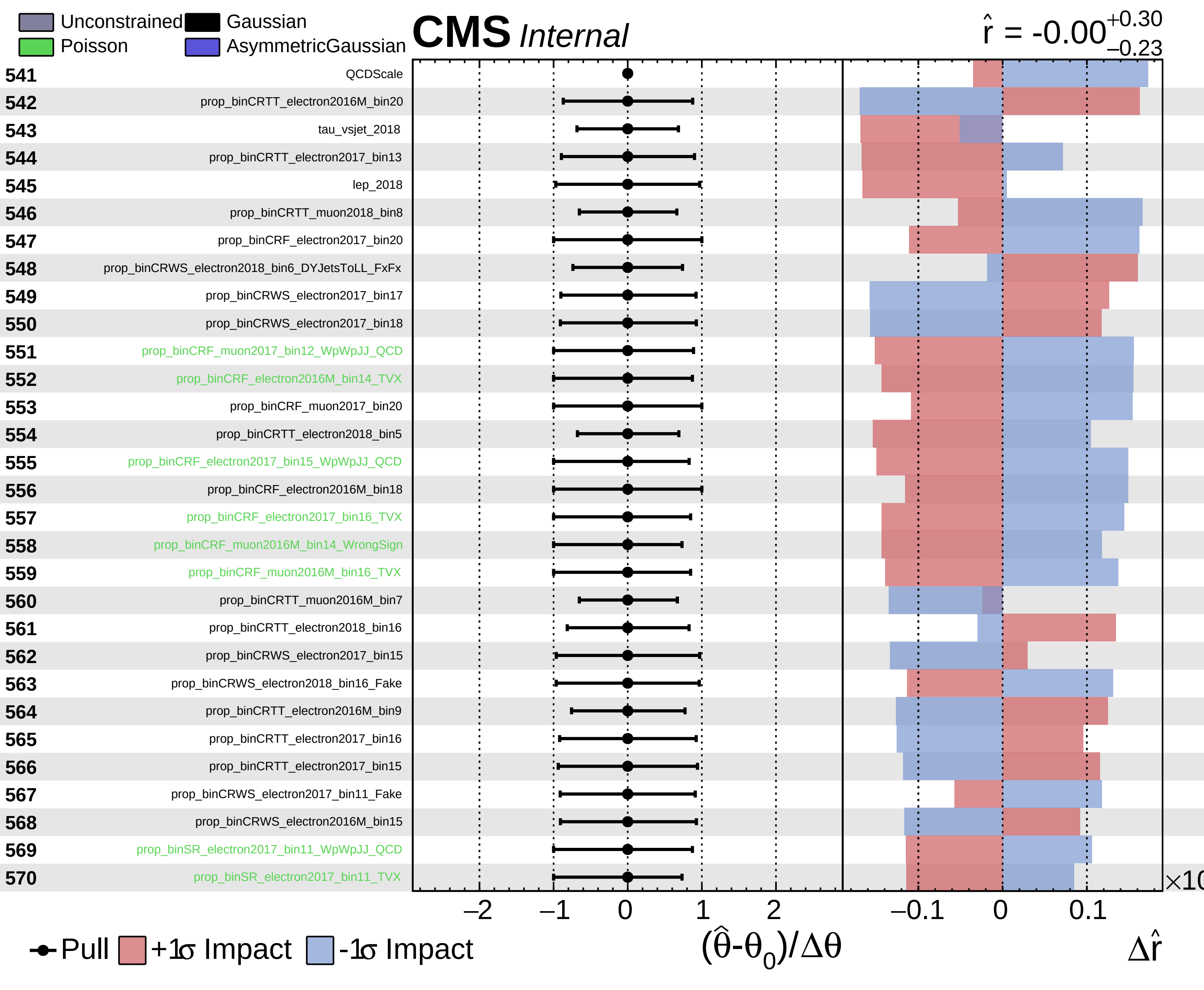


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
451 prop_binSR_muon2017_bin13_WpWpJJ_QCD
452 prop_binCRWS_electron2018_bin15
453 prop_binSR_electron2017_bin14_WpWpJJ_QCD
454 prop_binCRF_muon2017_bin9
455 prop_binCRTT_muon2017_bin9
456 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
457 prop_binCRTT_muon2017_bin14
458 prop_binCRF_muon2018_bin19
459 prop_binCRTT_electron2018_bin8
460 prop_binCRWS_electron2018_bin10
461 prop_binCRTT_electron2017_bin20
462 prop_binCRTT_muon2016M_bin11
463 prop_binCRTT_muon2017_bin8
464 prop_binCRTT_electron2017_bin12
465 prop_binCRF_muon2017_bin5
466 prop_binSR_electron2016M_bin5
467 prop_binCRF_electron2017_bin8
468 prop_binCRF_electron2016M_bin10
469 prop_binSR_electron2016M_bin3_Fake
470 prop_binCRWS_electron2018_bin8_Fake
471 prop_binSR_electron2017_bin19_WrongSign
472 prop_binCRF_muon2016M_bin5
473 prop_binCRTT_muon2016M_bin1
474 prop_binCRTT_muon2016M_bin5
475 prop_binCRWS_electron2016M_bin9_Fake
476 prop_binCRTT_electron2017_bin19
477 prop_binSR_electron2017_bin11_TTTo2L2Nu
478 tau_vsjet_2017
479 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
480 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

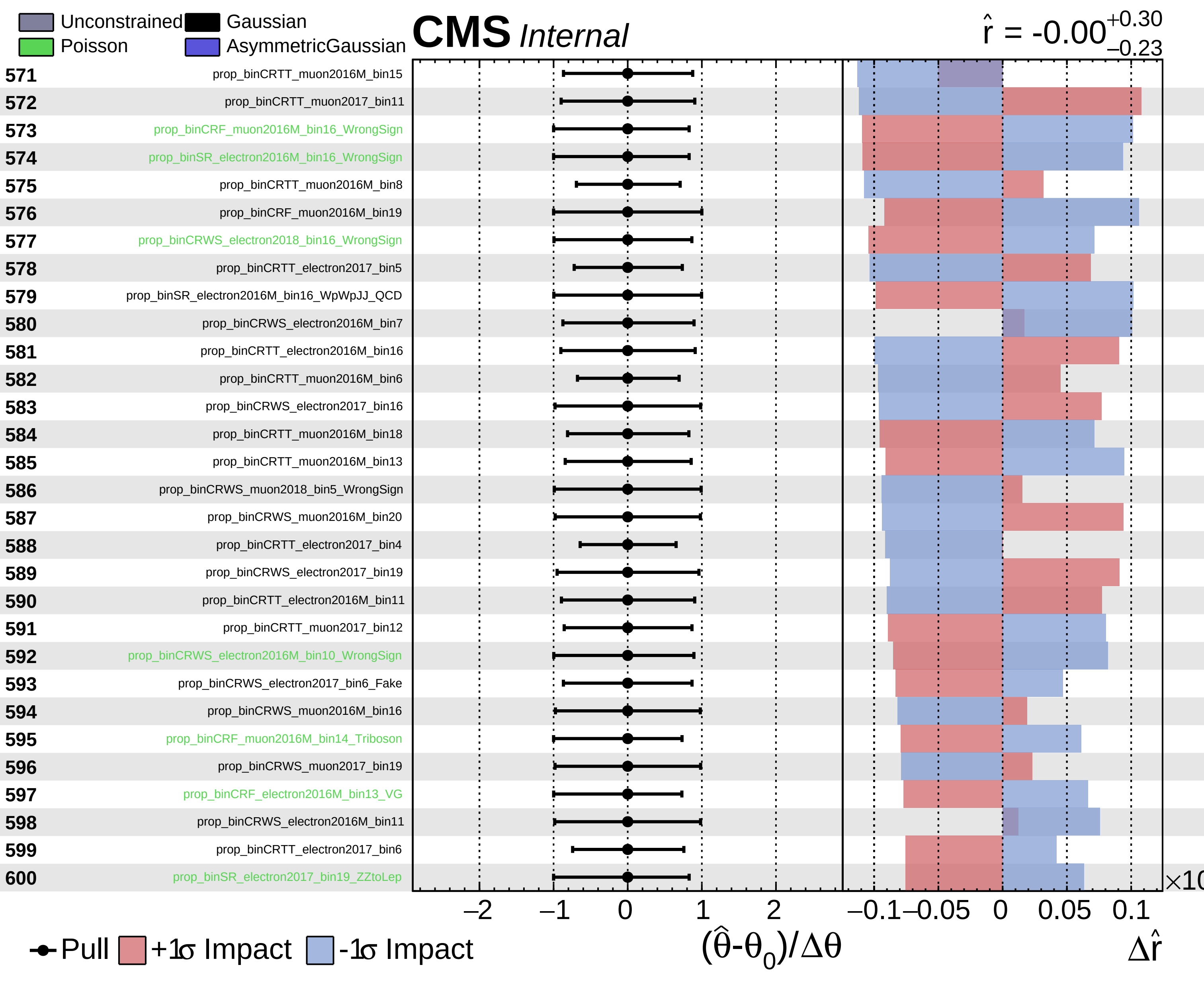

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
481 prop_binCRWS_muon2016M_bin19
482 prop_binCRWS_electron2018_bin19
483 prop_binCRF_muon2018_bin16_TVX
484 prop_binCRTT_muon2017_bin3
485 prop_binCRWS_electron2018_bin2
486 prop_binCRF_electron2017_bin18
487 prop_binCRTT_muon2017_bin16
488 prop_binCRTT_electron2018_bin12
489 prop_binCRF_muon2017_bin12_TVX
490 prop_binCRTT_muon2016M_bin17
491 prop_binSR_electron2017_bin19_Triboson
492 prop_binSR_muon2017_bin5
493 prop_binCRF_electron2017_bin19
494 prop_binCRF_electron2017_bin10
495 prop_binCRWS_electron2018_bin6_Fake
496 prop_binCRWS_electron2018_bin18
497 prop_binCRTT_electron2018_bin17
498 prop_binCRF_electron2017_bin16_WrongSign
499 prop_binCRF_muon2018_bin16_WrongSign
500 prop_binCRTT_muon2018_bin10
501 prop_binCRTT_electron2018_bin19
502 prop_binCRTT_muon2018_bin15
503 prop_binCRF_electron2016M_bin9
504 prop_binCRF_electron2018_bin7
505 tau_vsjet_2016M
506 prop_binCRF_electron2016M_bin20
507 prop_binCRTT_muon2018_bin19
508 prop_binCRWS_muon2016M_bin12
509 prop_binCRWS_electron2017_bin8
510 prop_binCRF_muon2016M_bin14_WpWpJJ_QCD
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



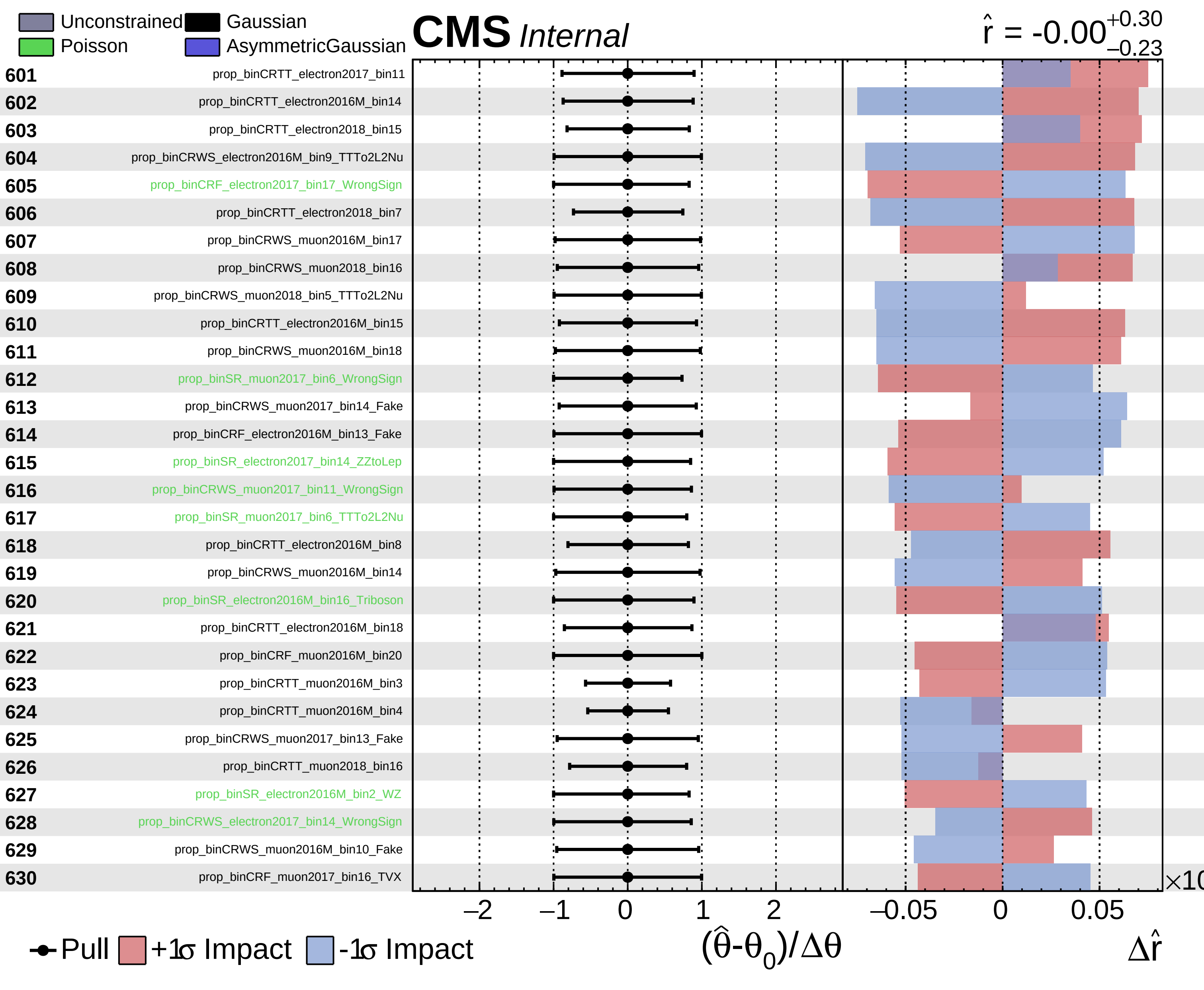
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
511 prop_binCRF_muon2017_bin18
512 prop_binCRWS_muon2016M_bin13
513 prop_binCRF_muon2018_bin20
514 prop_binCRTT_muon2017_bin15
515 prop_binCRTT_muon2018_bin2
516 prop_binCRWS_muon2016M_bin15
517 prop_binCRWS_muon2017_bin11_Fake
518 prop_binCRF_electron2016M_bin7
519 prop_binCRTT_muon2016M_bin10
520 prop_binSR_electron2017_bin14_Triboson
521 prop_binCRF_muon2018_bin16_WpWpJJ_QCD
522 prop_binSR_electron2016M_bin16_TVX
523 prop_binCRTT_muon2017_bin7
524 prop_binCRTT_electron2017_bin14
525 prop_binCRF_muon2017_bin12_WrongSign
526 prop_binCRTT_electron2016M_bin12
527 puID
528 prop_binCRTT_electron2016M_bin5
529 prop_binCRWS_electron2017_bin14_Fake
530 prop_binSR_electron2017_bin12_TVX
531 prop_binCRF_electron2016M_bin8
532 prop_binCRTT_electron2018_bin6
533 prop_binCRTT_muon2018_bin13
534 prop_binCRTT_muon2016M_bin16
535 prop_binCRTT_electron2017_bin17
536 prop_binCRWS_muon2017_bin7_Fake
537 prop_binCRTT_electron2016M_bin13
538 prop_binSR_electron2017_bin12_Triboson
539 prop_binCRTT_electron2017_bin7
540 prop_binCRWS_muon2017_bin17
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



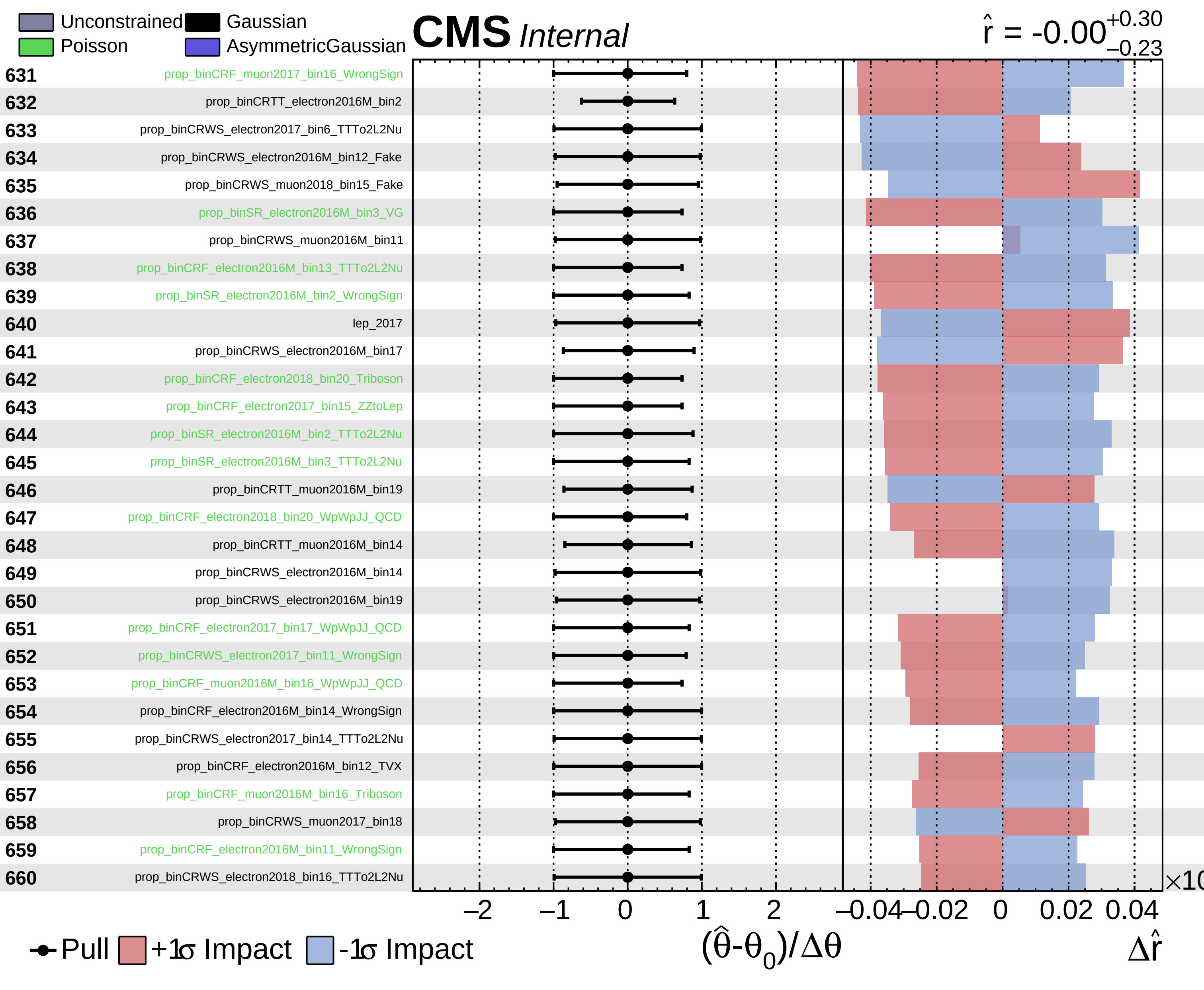
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
541 QCDScale
542 prop_binCRTT_electron2016M_bin20
543 tau_vsjet_2018
544 prop_binCRTT_electron2017_bin13
545 lep_2018
546 prop_binCRTT_muon2018_bin8
547 prop_binCRF_electron2017_bin20
548 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
549 prop_binCRWS_electron2017_bin17
550 prop_binCRWS_electron2017_bin18
551 prop_binCRF_muon2017_bin12_WpWpJJ_QCD
552 prop_binCRF_electron2016M_bin14_TVX
553 prop_binCRF_muon2017_bin20
554 prop_binCRTT_electron2018_bin5
555 prop_binCRF_electron2017_bin15_WpWpJJ_QCD
556 prop_binCRF_electron2016M_bin18
557 prop_binCRF_electron2017_bin16_TVX
558 prop_binCRF_muon2016M_bin14_WrongSign
559 prop_binCRF_muon2016M_bin16_TVX
560 prop_binCRTT_muon2016M_bin7
561 prop_binCRTT_electron2018_bin16
562 prop_binCRWS_electron2017_bin15
563 prop_binCRWS_electron2018_bin16_Fake
564 prop_binCRTT_electron2016M_bin9
565 prop_binCRTT_electron2017_bin16
566 prop_binCRTT_electron2017_bin15
567 prop_binCRWS_electron2017_bin11_Fake
568 prop_binCRWS_electron2016M_bin15
569 prop_binSR_electron2017_bin11_WpWpJJ_QCD
570 prop_binSR_electron2017_bin11_TVX
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



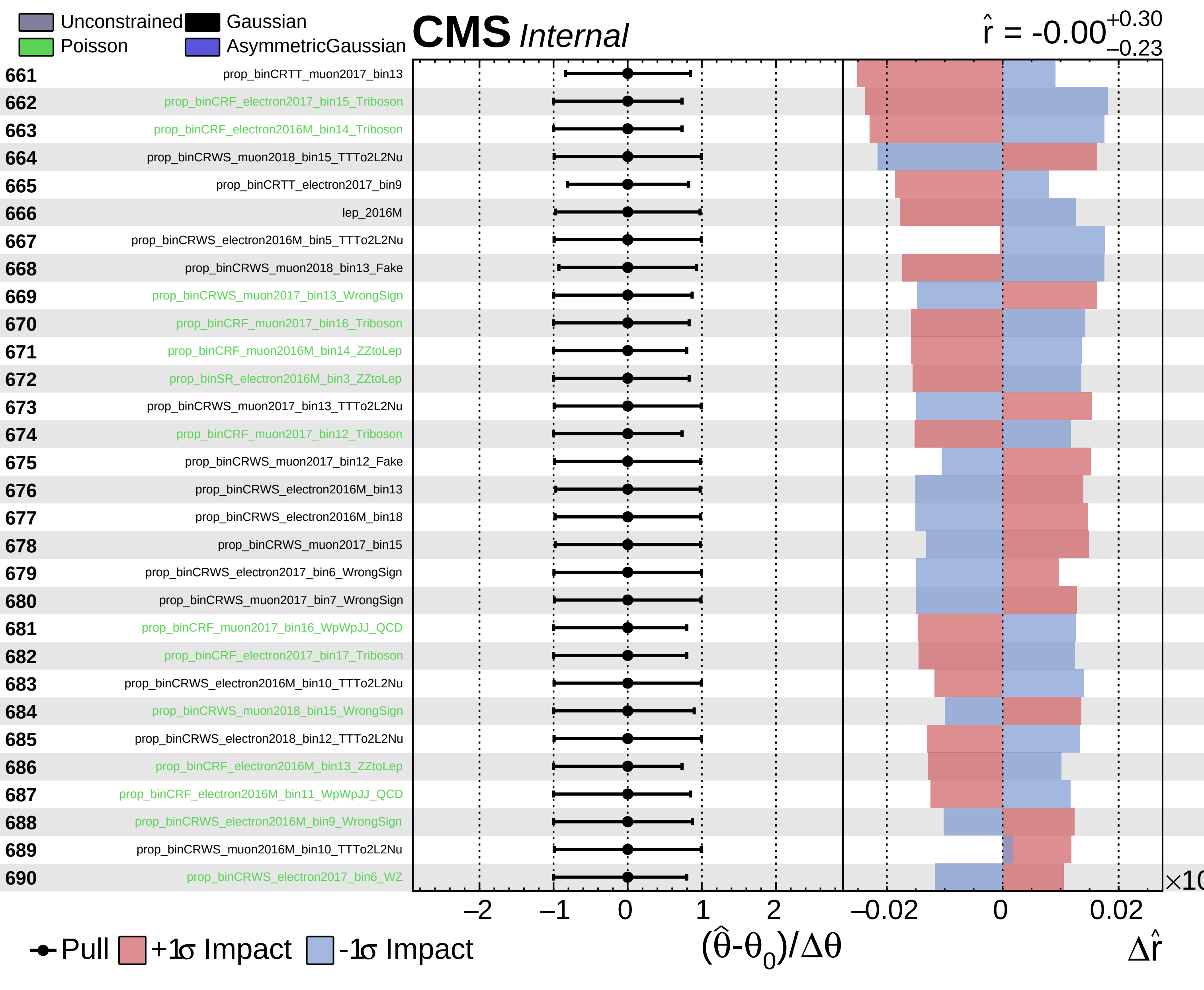
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
571 prop_binCRTT_muon2016M_bin15
572 prop_binCRTT_muon2017_bin11
573 prop_binCRF_muon2016M_bin16_WrongSign
574 prop_binSR_electron2016M_bin16_WrongSign
575 prop_binCRTT_muon2016M_bin8
576 prop_binCRF_muon2016M_bin19
577 prop_binCRWS_electron2018_bin16_WrongSign
578 prop_binCRTT_electron2017_bin5
579 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
580 prop_binCRWS_electron2016M_bin7
581 prop_binCRTT_electron2016M_bin16
582 prop_binCRTT_muon2016M_bin6
583 prop_binCRWS_electron2017_bin16
584 prop_binCRTT_muon2016M_bin18
585 prop_binCRTT_muon2016M_bin13
586 prop_binCRWS_muon2018_bin5_WrongSign
587 prop_binCRWS_muon2016M_bin20
588 prop_binCRTT_electron2017_bin4
589 prop_binCRWS_electron2017_bin19
590 prop_binCRTT_electron2016M_bin11
591 prop_binCRTT_muon2017_bin12
592 prop_binCRWS_electron2016M_bin10_WrongSign
593 prop_binCRWS_electron2017_bin6_Fake
594 prop_binCRWS_muon2016M_bin16
595 prop_binCRF_muon2016M_bin14_Triboson
596 prop_binCRWS_muon2017_bin19
597 prop_binCRF_electron2016M_bin13_VG
598 prop_binCRWS_electron2016M_bin11
599 prop_binCRTT_electron2017_bin6
600 prop_binSR_electron2017_bin19_ZZtoLep
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



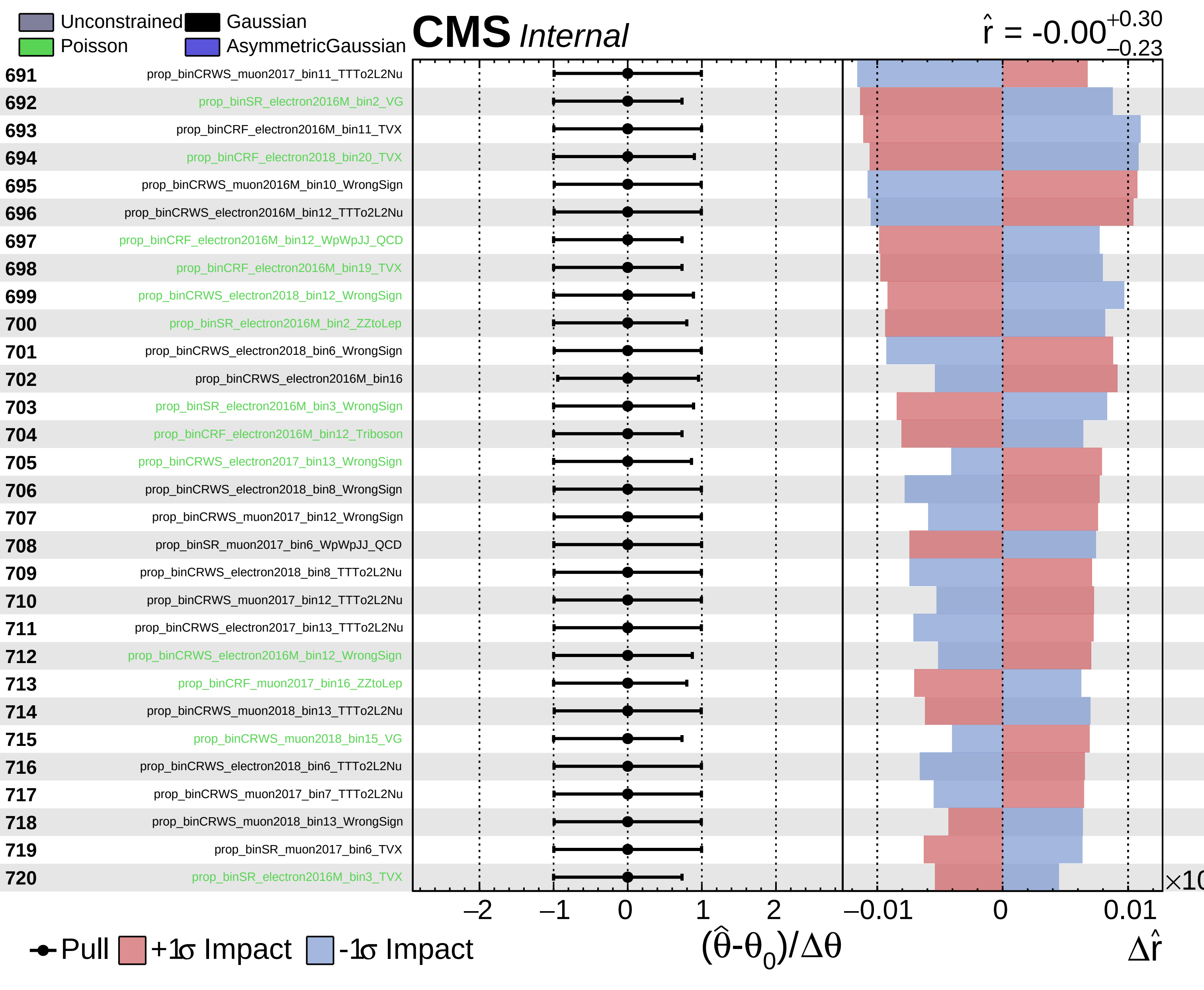
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
601 prop_binCRTT_electron2017_bin11
602 prop_binCRTT_electron2016M_bin14
603 prop_binCRTT_electron2018_bin15
604 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
605 prop_binCRF_electron2017_bin17_WrongSign
606 prop_binCRTT_electron2018_bin7
607 prop_binCRWS_muon2016M_bin17
608 prop_binCRWS_muon2018_bin16
609 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
610 prop_binCRTT_electron2016M_bin15
611 prop_binCRWS_muon2016M_bin18
612 prop_binSR_muon2017_bin6_WrongSign
613 prop_binCRWS_muon2017_bin14_Fake
614 prop_binCRF_electron2016M_bin13_Fake
615 prop_binSR_electron2017_bin14_ZZtoLep
616 prop_binCRWS_muon2017_bin11_WrongSign
617 prop_binSR_muon2017_bin6_TTTo2L2Nu
618 prop_binCRTT_electron2016M_bin8
619 prop_binCRWS_muon2016M_bin14
620 prop_binSR_electron2016M_bin16_Triboson
621 prop_binCRTT_electron2016M_bin18
622 prop_binCRF_muon2016M_bin20
623 prop_binCRTT_muon2016M_bin3
624 prop_binCRTT_muon2016M_bin4
625 prop_binCRWS_muon2017_bin13_Fake
626 prop_binCRTT_muon2018_bin16
627 prop_binSR_electron2016M_bin2_WZ
628 prop_binCRWS_electron2017_bin14_WrongSign
629 prop_binCRWS_muon2016M_bin10_Fake
630 prop_binCRF_muon2017_bin16_TVX
−2 −1 0 1 2
−0.05 0 0.05
$\times 10$
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



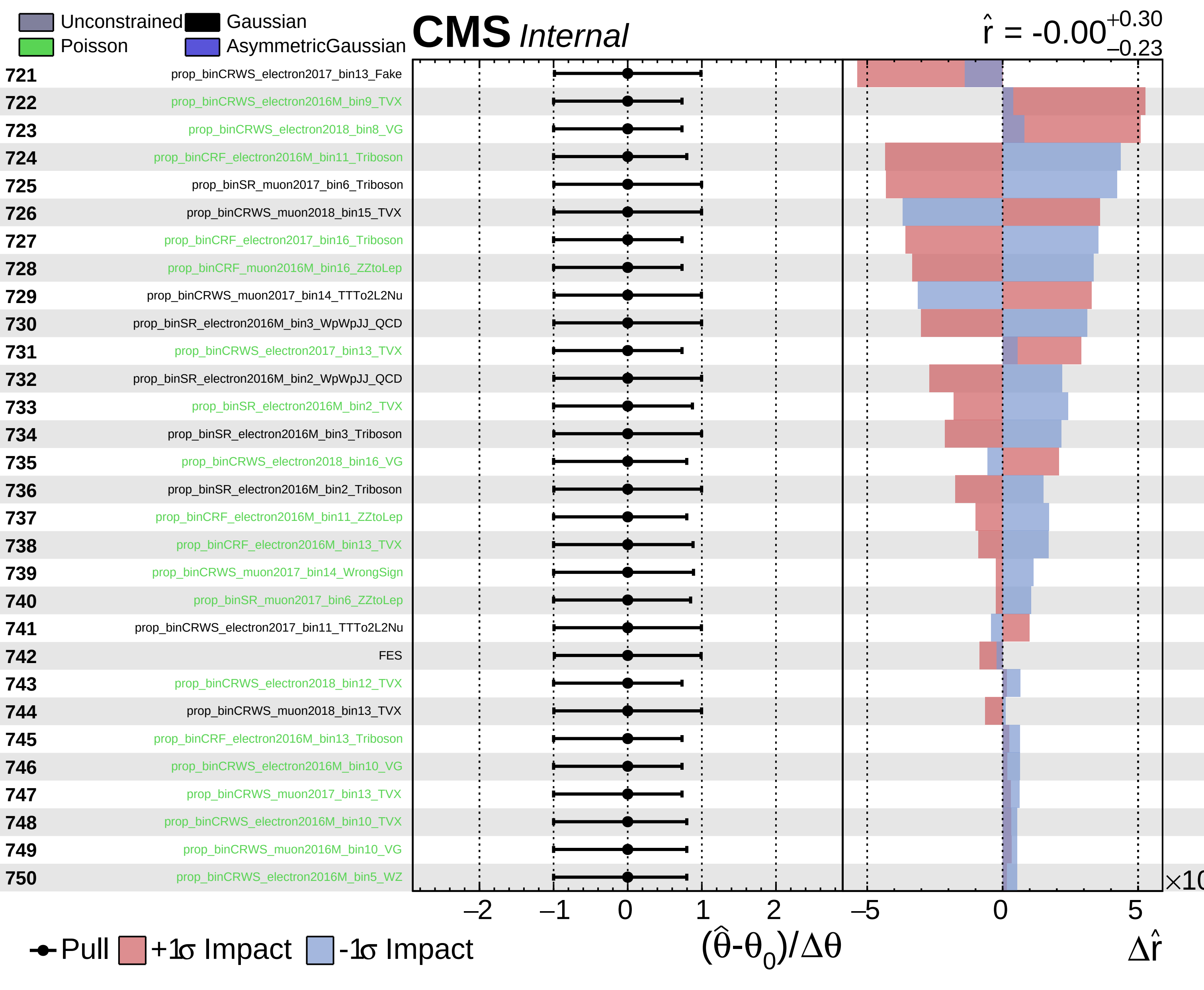
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
631 prop_binCRF_muon2017_bin16_WrongSign
632 prop_binCRTT_electron2016M_bin2
633 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
634 prop_binCRWS_electron2016M_bin12_Fake
635 prop_binCRWS_muon2018_bin15_Fake
636 prop_binSR_electron2016M_bin3_VG
637 prop_binCRWS_muon2016M_bin11
638 prop_binCRF_electron2016M_bin13_TTTo2L2Nu
639 prop_binSR_electron2016M_bin2_WrongSign
640 lep_2017
641 prop_binCRWS_electron2016M_bin17
642 prop_binCRF_electron2018_bin20_Triboson
643 prop_binCRF_electron2017_bin15_ZZtoLep
644 prop_binSR_electron2016M_bin2_TTTo2L2Nu
645 prop_binSR_electron2016M_bin3_TTTo2L2Nu
646 prop_binCRTT_muon2016M_bin19
647 prop_binCRF_electron2018_bin20_WpWpJJ_QCD
648 prop_binCRTT_muon2016M_bin14
649 prop_binCRWS_electron2016M_bin14
650 prop_binCRWS_electron2016M_bin19
651 prop_binCRF_electron2017_bin17_WpWpJJ_QCD
652 prop_binCRWS_electron2017_bin11_WrongSign
653 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
654 prop_binCRF_electron2016M_bin14_WrongSign
655 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
656 prop_binCRF_electron2016M_bin12_TVX
657 prop_binCRF_muon2016M_bin16_Triboson
658 prop_binCRWS_muon2017_bin18
659 prop_binCRF_electron2016M_bin11_WrongSign
660 prop_binCRWS_electron2018_bin16_TTTo2L2Nu
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



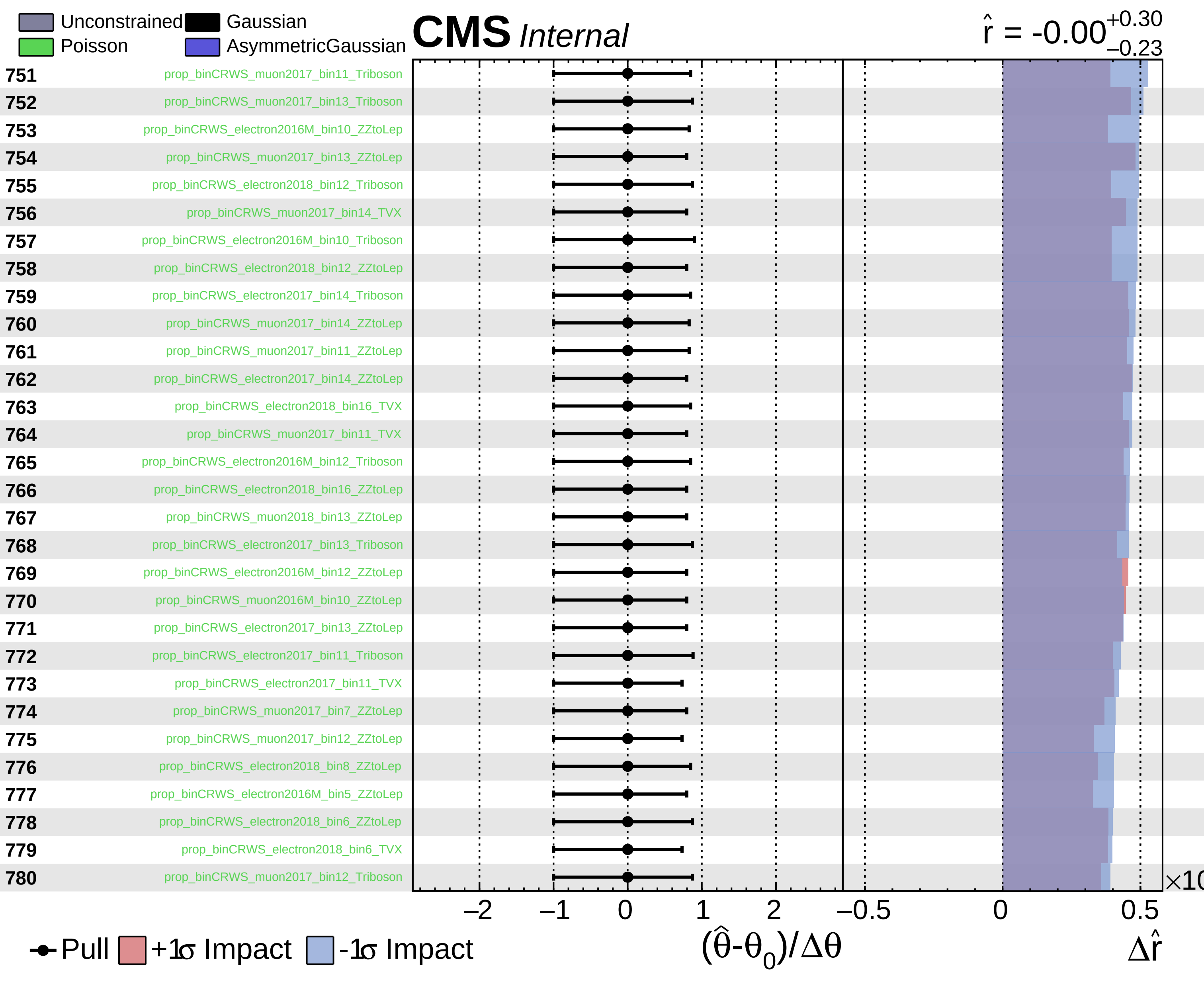
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
661 prop_binCRTT_muon2017_bin13
662 prop_binCRF_electron2017_bin15_Triboson
663 prop_binCRF_electron2016M_bin14_Triboson
664 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
665 prop_binCRTT_electron2017_bin9
666 lep_2016M
667 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
668 prop_binCRWS_muon2018_bin13_Fake
669 prop_binCRWS_muon2017_bin13_WrongSign
670 prop_binCRF_muon2017_bin16_Triboson
671 prop_binCRF_muon2016M_bin14_ZZtoLep
672 prop_binSR_electron2016M_bin3_ZZtoLep
673 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
674 prop_binCRF_muon2017_bin12_Triboson
675 prop_binCRWS_muon2017_bin12_Fake
676 prop_binCRWS_electron2016M_bin13
677 prop_binCRWS_electron2016M_bin18
678 prop_binCRWS_muon2017_bin15
679 prop_binCRWS_electron2017_bin6_WrongSign
680 prop_binCRWS_muon2017_bin7_WrongSign
681 prop_binCRF_muon2017_bin16_WpWpJJ_QCD
682 prop_binCRF_electron2017_bin17_Triboson
683 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
684 prop_binCRWS_muon2018_bin15_WrongSign
685 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
686 prop_binCRF_electron2016M_bin13_ZZtoLep
687 prop_binCRF_electron2016M_bin11_WpWpJJ_QCD
688 prop_binCRWS_electron2016M_bin9_WrongSign
689 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
690 prop_binCRWS_electron2017_bin6_WZ
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



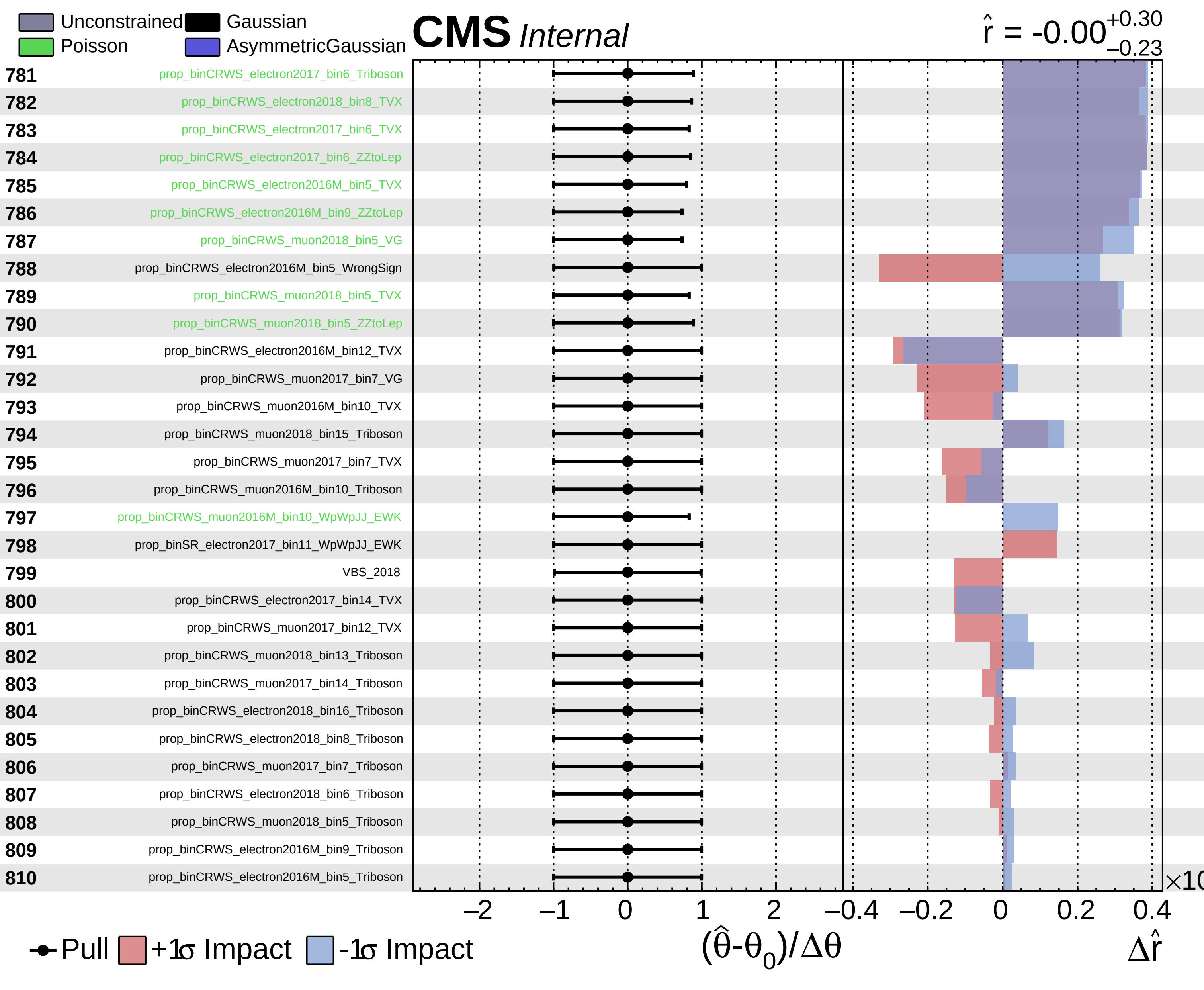
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
691 prop_binCRWS_muon2017_bin11_TTTo2L2Nu
692 prop_binSR_electron2016M_bin2_VG
693 prop_binCRF_electron2016M_bin11_TVX
694 prop_binCRF_electron2018_bin20_TVX
695 prop_binCRWS_muon2016M_bin10_WrongSign
696 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
697 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
698 prop_binCRF_electron2016M_bin19_TVX
699 prop_binCRWS_electron2018_bin12_WrongSign
700 prop_binSR_electron2016M_bin2_ZZtoLep
701 prop_binCRWS_electron2018_bin6_WrongSign
702 prop_binCRWS_electron2016M_bin16
703 prop_binSR_electron2016M_bin3_WrongSign
704 prop_binCRF_electron2016M_bin12_Triboson
705 prop_binCRWS_electron2017_bin13_WrongSign
706 prop_binCRWS_electron2018_bin8_WrongSign
707 prop_binCRWS_muon2017_bin12_WrongSign
708 prop_binSR_muon2017_bin6_WpWpJJ_QCD
709 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
710 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
711 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
712 prop_binCRWS_electron2016M_bin12_WrongSign
713 prop_binCRF_muon2017_bin16_ZZtoLep
714 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
715 prop_binCRWS_muon2018_bin15_VG
716 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
717 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
718 prop_binCRWS_muon2018_bin13_WrongSign
719 prop_binSR_muon2017_bin6_TVX
720 prop_binSR_electron2016M_bin3_TVX
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



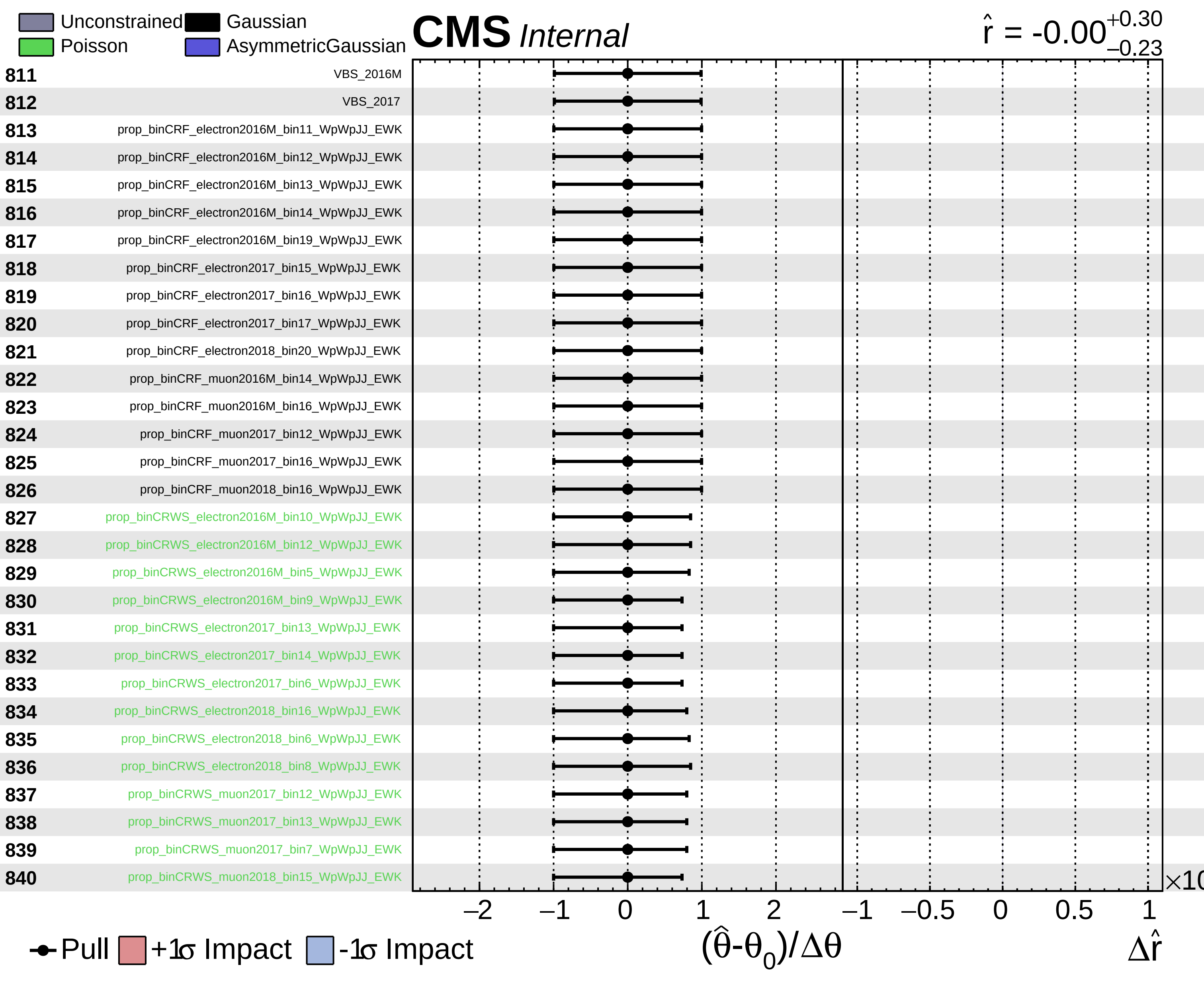
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
721 prop_binCRWS_electron2017_bin13_Fake
722 prop_binCRWS_electron2016M_bin9_TVX
723 prop_binCRWS_electron2018_bin8_VG
724 prop_binCRF_electron2016M_bin11_Triboson
725 prop_binSR_muon2017_bin6_Triboson
726 prop_binCRWS_muon2018_bin15_TVX
727 prop_binCRF_electron2017_bin16_Triboson
728 prop_binCRF_muon2016M_bin16_ZZtoLep
729 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
730 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
731 prop_binCRWS_electron2017_bin13_TVX
732 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
733 prop_binSR_electron2016M_bin2_TVX
734 prop_binSR_electron2016M_bin3_Triboson
735 prop_binCRWS_electron2018_bin16_VG
736 prop_binSR_electron2016M_bin2_Triboson
737 prop_binCRF_electron2016M_bin11_ZZtoLep
738 prop_binCRF_electron2016M_bin13_TVX
739 prop_binCRWS_muon2017_bin14_WrongSign
740 prop_binSR_muon2017_bin6_ZZtoLep
741 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
742 FES
743 prop_binCRWS_electron2018_bin12_TVX
744 prop_binCRWS_muon2018_bin13_TVX
745 prop_binCRF_electron2016M_bin13_Triboson
746 prop_binCRWS_electron2016M_bin10_VG
747 prop_binCRWS_muon2017_bin13_TVX
748 prop_binCRWS_electron2016M_bin10_TVX
749 prop_binCRWS_muon2016M_bin10_VG
750 prop_binCRWS_electron2016M_bin5_WZ
−2 −1 0 1 2
−5 0 5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



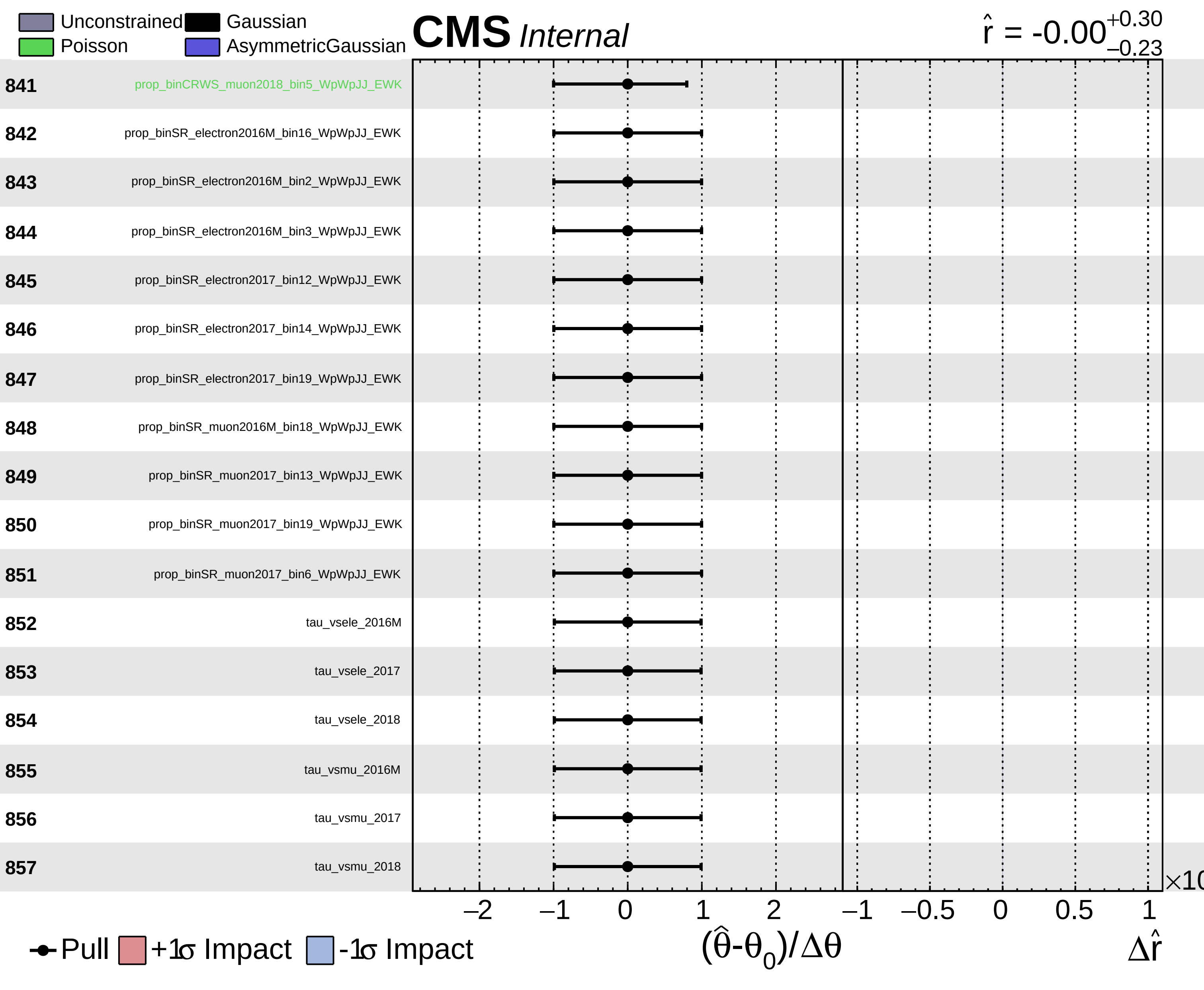
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
751 prop_binCRWS_muon2017_bin11_Triboson
752 prop_binCRWS_muon2017_bin13_Triboson
753 prop_binCRWS_electron2016M_bin10_ZZtoLep
754 prop_binCRWS_muon2017_bin13_ZZtoLep
755 prop_binCRWS_electron2018_bin12_Triboson
756 prop_binCRWS_muon2017_bin14_TVX
757 prop_binCRWS_electron2016M_bin10_Triboson
758 prop_binCRWS_electron2018_bin12_ZZtoLep
759 prop_binCRWS_electron2017_bin14_Triboson
760 prop_binCRWS_muon2017_bin14_ZZtoLep
761 prop_binCRWS_muon2017_bin11_ZZtoLep
762 prop_binCRWS_electron2017_bin14_ZZtoLep
763 prop_binCRWS_electron2018_bin16_TVX
764 prop_binCRWS_muon2017_bin11_TVX
765 prop_binCRWS_electron2016M_bin12_Triboson
766 prop_binCRWS_electron2018_bin16_ZZtoLep
767 prop_binCRWS_muon2018_bin13_ZZtoLep
768 prop_binCRWS_electron2017_bin13_Triboson
769 prop_binCRWS_electron2016M_bin12_ZZtoLep
770 prop_binCRWS_muon2016M_bin10_ZZtoLep
771 prop_binCRWS_electron2017_bin13_ZZtoLep
772 prop_binCRWS_electron2017_bin11_Triboson
773 prop_binCRWS_electron2017_bin11_TVX
774 prop_binCRWS_muon2017_bin7_ZZtoLep
775 prop_binCRWS_muon2017_bin12_ZZtoLep
776 prop_binCRWS_electron2018_bin8_ZZtoLep
777 prop_binCRWS_electron2016M_bin5_ZZtoLep
778 prop_binCRWS_electron2018_bin6_ZZtoLep
779 prop_binCRWS_electron2018_bin6_TVX
780 prop_binCRWS_muon2017_bin12_Triboson
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
781 prop_binCRWS_electron2017_bin6_Triboson
782 prop_binCRWS_electron2018_bin8_TVX
783 prop_binCRWS_electron2017_bin6_TVX
784 prop_binCRWS_electron2017_bin6_ZZtoLep
785 prop_binCRWS_electron2016M_bin5_TVX
786 prop_binCRWS_electron2016M_bin9_ZZtoLep
787 prop_binCRWS_muon2018_bin5_VG
788 prop_binCRWS_electron2016M_bin5_WrongSign
789 prop_binCRWS_muon2018_bin5_TVX
790 prop_binCRWS_muon2018_bin5_ZZtoLep
791 prop_binCRWS_electron2016M_bin12_TVX
792 prop_binCRWS_muon2017_bin7_VG
793 prop_binCRWS_muon2016M_bin10_TVX
794 prop_binCRWS_muon2018_bin15_Triboson
795 prop_binCRWS_muon2017_bin7_TVX
796 prop_binCRWS_muon2016M_bin10_Triboson
797 prop_binCRWS_muon2016M_bin10_WpWpJJ_EWK
798 prop_binSR_electron2017_bin11_WpWpJJ_EWK
799 VBS_2018
800 prop_binCRWS_electron2017_bin14_TVX
801 prop_binCRWS_muon2017_bin12_TVX
802 prop_binCRWS_muon2018_bin13_Triboson
803 prop_binCRWS_muon2017_bin14_Triboson
804 prop_binCRWS_electron2018_bin16_Triboson
805 prop_binCRWS_electron2018_bin8_Triboson
806 prop_binCRWS_muon2017_bin7_Triboson
807 prop_binCRWS_electron2018_bin6_Triboson
808 prop_binCRWS_muon2018_bin5_Triboson
809 prop_binCRWS_electron2016M_bin9_Triboson
810 prop_binCRWS_electron2016M_bin5_Triboson
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.30 −0.23
811 VBS_2016M
812 VBS_2017
813 prop_binCRF_electron2016M_bin11_WpWpJJ_EWK
814 prop_binCRF_electron2016M_bin12_WpWpJJ_EWK
815 prop_binCRF_electron2016M_bin13_WpWpJJ_EWK
816 prop_binCRF_electron2016M_bin14_WpWpJJ_EWK
817 prop_binCRF_electron2016M_bin19_WpWpJJ_EWK
818 prop_binCRF_electron2017_bin15_WpWpJJ_EWK
819 prop_binCRF_electron2017_bin16_WpWpJJ_EWK
820 prop_binCRF_electron2017_bin17_WpWpJJ_EWK
821 prop_binCRF_electron2018_bin20_WpWpJJ_EWK
822 prop_binCRF_muon2016M_bin14_WpWpJJ_EWK
823 prop_binCRF_muon2016M_bin16_WpWpJJ_EWK
824 prop_binCRF_muon2017_bin12_WpWpJJ_EWK
825 prop_binCRF_muon2017_bin16_WpWpJJ_EWK
826 prop_binCRF_muon2018_bin16_WpWpJJ_EWK
827 prop_binCRWS_electron2016M_bin10_WpWpJJ_EWK
828 prop_binCRWS_electron2016M_bin12_WpWpJJ_EWK
829 prop_binCRWS_electron2016M_bin5_WpWpJJ_EWK
830 prop_binCRWS_electron2016M_bin9_WpWpJJ_EWK
831 prop_binCRWS_electron2017_bin13_WpWpJJ_EWK
832 prop_binCRWS_electron2017_bin14_WpWpJJ_EWK
833 prop_binCRWS_electron2017_bin6_WpWpJJ_EWK
834 prop_binCRWS_electron2018_bin16_WpWpJJ_EWK
835 prop_binCRWS_electron2018_bin6_WpWpJJ_EWK
836 prop_binCRWS_electron2018_bin8_WpWpJJ_EWK
837 prop_binCRWS_muon2017_bin12_WpWpJJ_EWK
838 prop_binCRWS_muon2017_bin13_WpWpJJ_EWK
839 prop_binCRWS_muon2017_bin7_WpWpJJ_EWK
840 prop_binCRWS_muon2018_bin15_WpWpJJ_EWK
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.30}_{-0.23}$
841 prop_binCRWS_muon2018_bin5_WpWpJJ_EWK
842 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
843 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
844 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
845 prop_binSR_electron2017_bin12_WpWpJJ_EWK
846 prop_binSR_electron2017_bin14_WpWpJJ_EWK
847 prop_binSR_electron2017_bin19_WpWpJJ_EWK
848 prop_binSR_muon2016M_bin18_WpWpJJ_EWK
849 prop_binSR_muon2017_bin13_WpWpJJ_EWK
850 prop_binSR_muon2017_bin19_WpWpJJ_EWK
851 prop_binSR_muon2017_bin6_WpWpJJ_EWK
852 tau_vsele_2016M
853 tau_vsele_2017
854 tau_vsele_2018
855 tau_vsmu_2016M
856 tau_vsmu_2017
857 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$